

Physical-World Observation

for Personal AI Agent Personalization



Personal AI Agents are Here

A shift from reactive chatbots to intelligent systems with **continuous memory** that **deeply understand** you and **proactively act** on your behalf.

AI Assistant (Chatbot)



Reactive

Waits for your prompt to act. Transactional, turn-based back-and-forth interactions.



Isolated Sessions

Each interaction starts fresh. Needs context re-explained repeatedly.



General Knowledge

Provides answers based on broad training data. Uniform output across all users.

TLDR AI assistants are focused on transient, immediate tasks.

Personal AI Agent



Proactive

Anticipates needs, initiates actions, and handles background tasks autonomously.



Continuous Context

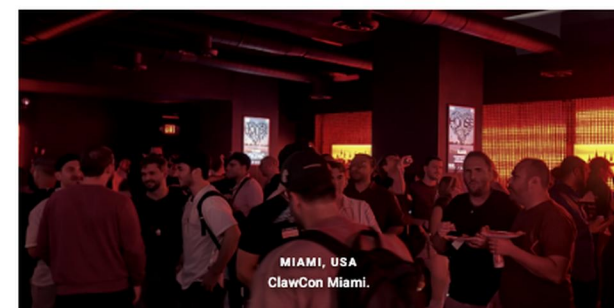
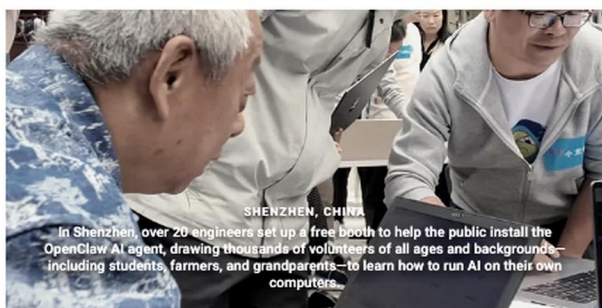
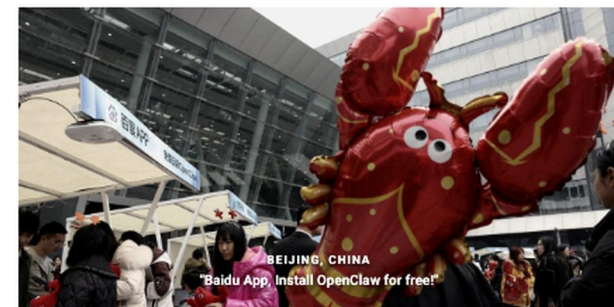
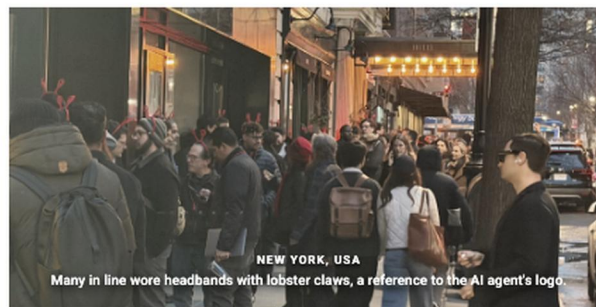
Builds a lasting understanding of your life, habits, workflows, and preferences.



Deeply Personalized

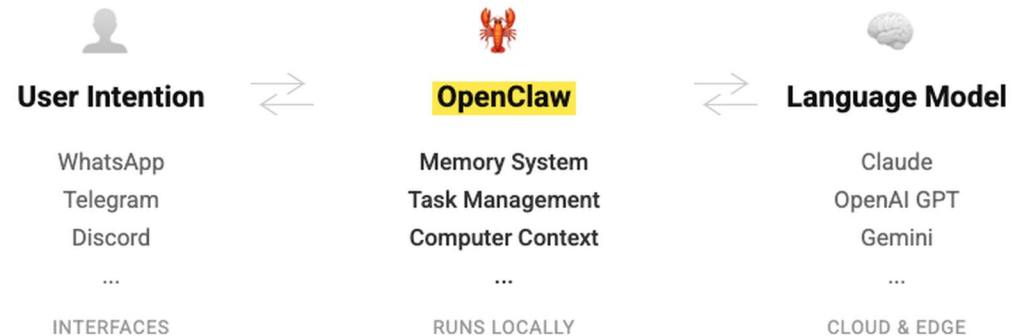
Tailored uniquely to you, seamlessly adapting to your specific context and requirements.

TLDR Personal AI Agents track long-term work and projects, managing various life aspects simultaneously while never forgetting who you are—enabling a fundamentally different relationship and interaction paradigm.



What is 🦞 OpenClaw?

An **open-source autonomous AI agent framework** designed to function as a personal AI assistant that executes complex workflows.



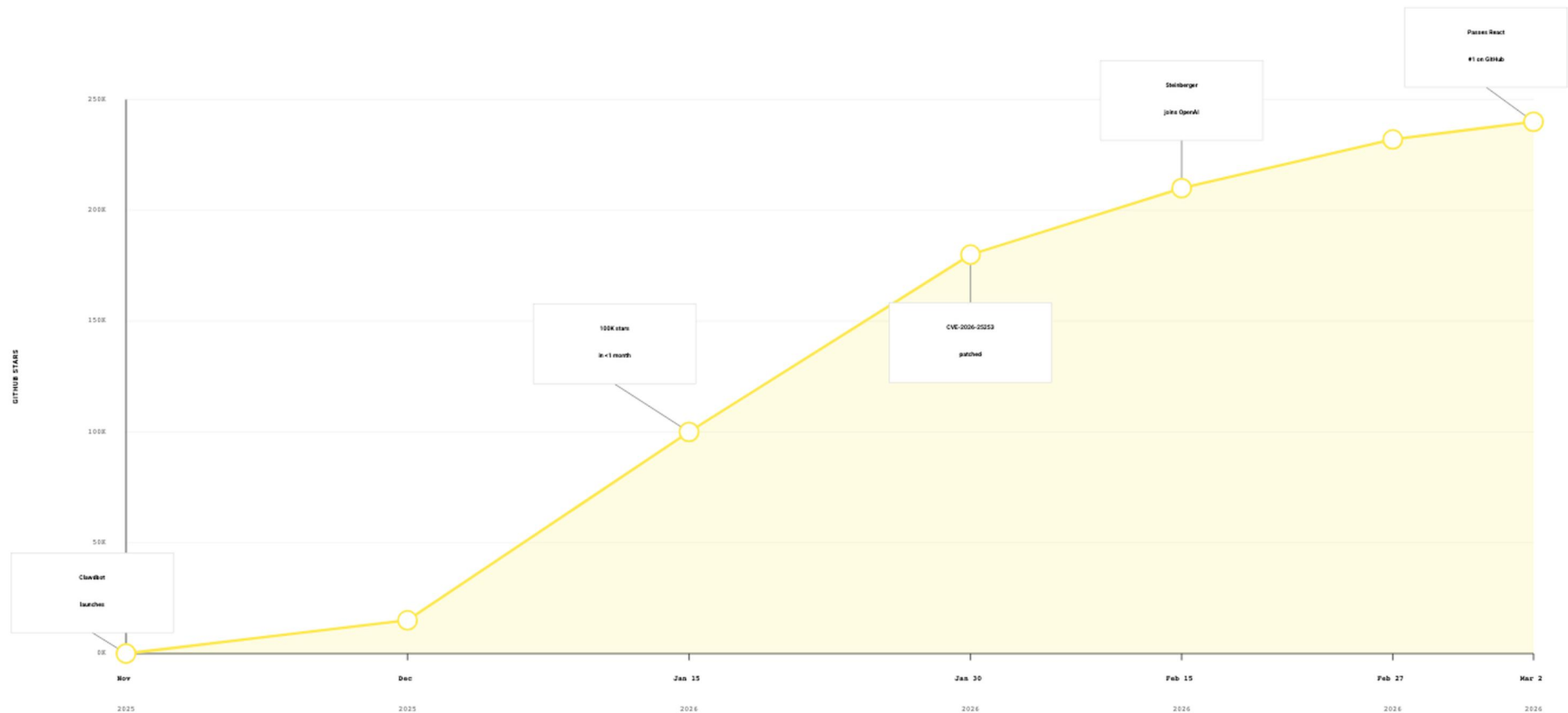


 **OpenClaw** isn't just an iteration; it's an unprecedented global cognitive layer spreading at hyper-velocity.

Source: declawed.io (Snapshot: 2026-04-06)
Estimated global deployment distribution based on public exposure data



OpenClaw: Fastest-Growing Open Source Project in GitHub History



Source: GitHub (Snapshot: 2026-03-02)
Historical growth data indexed from public repository markers

Defining Personalization

"Personalization is a process that changes the functionality, interface, information content, or distinctiveness of a system as to increase its personal relevance to an individual."

— Blom, J. (2000). *Personalization - a taxonomy*.

OUR GOAL Modifying the system so that the service or assistant **fits or exceeds the expectation**.

Normal System

-  **Pre-programmed**
Static rules and generic logic designed for the average user.
-  **Low Personalization**
Limited mostly to recommendation algorithms and targeted ads.
-  **Invisible**
Changes happen silently behind the scenes; hard to perceive.

TLDR One-size-fits-all approach with minimal dynamic adaptation.

AI Assistant (LLM)

-  **Probabilistic**
Complex, non-deterministic responses based on vast training distributions.
-  **Real-time Adaptation**
Can dynamically shift tone and output based on immediate human feedback.
-  **Evolving Models**
A spectrum of alignment techniques from pre-training to post-training.

TLDR Highly flexible systems capable of aligning deeply with individual feedback.

Human Interaction

-  **Constant Observation**
Unconsciously reading cues, context, and subtle emotional states.
-  **Immediate Loop**
Instantly sensing misunderstandings and adjusting explanations on the fly.
-  **Mutual Alignment**
A two-way street of continuous empathy and shared understanding.

TLDR The ultimate gold standard for personalized, context-aware communication.

DEGREE OF PERSONALIZATION →

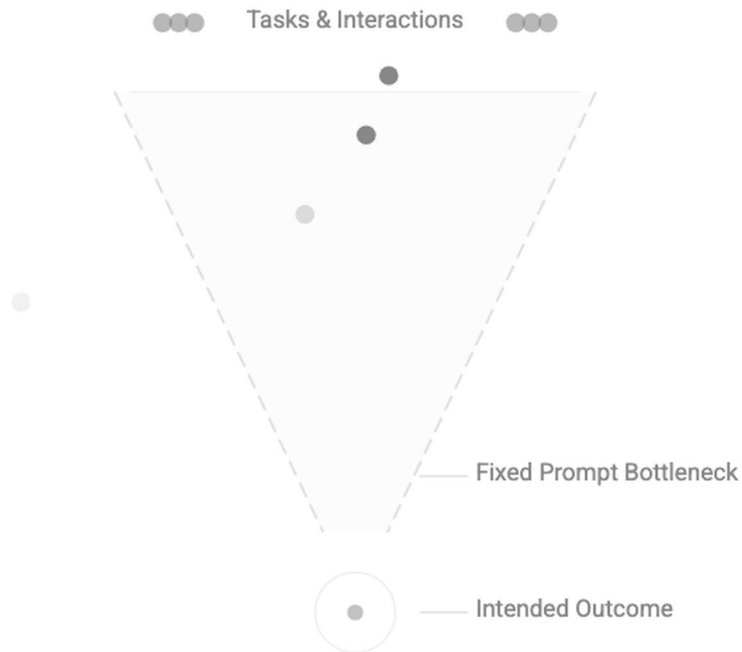
Why Personalization Matters

Generic Agent: Same behavior for everyone | Personalized Agent: Adapts, learns, evolves with you.

Personalization == Effectiveness

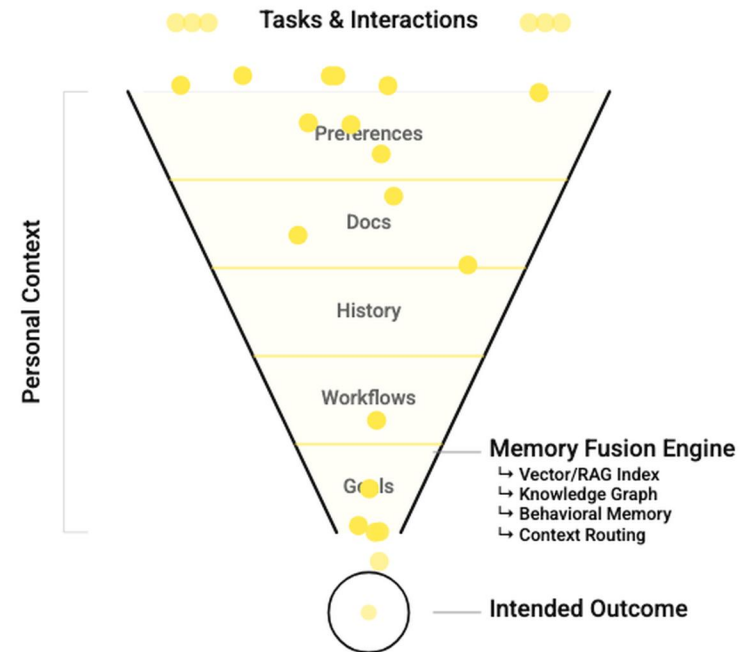
Agents That Don't Know You

Context leaks during the transaction. Only a generic prompt hits the target.



Agents That Know You

Captures and fuses your continuous context for hyper-relevant action.



THE BLIND SPOT



The Two Worlds

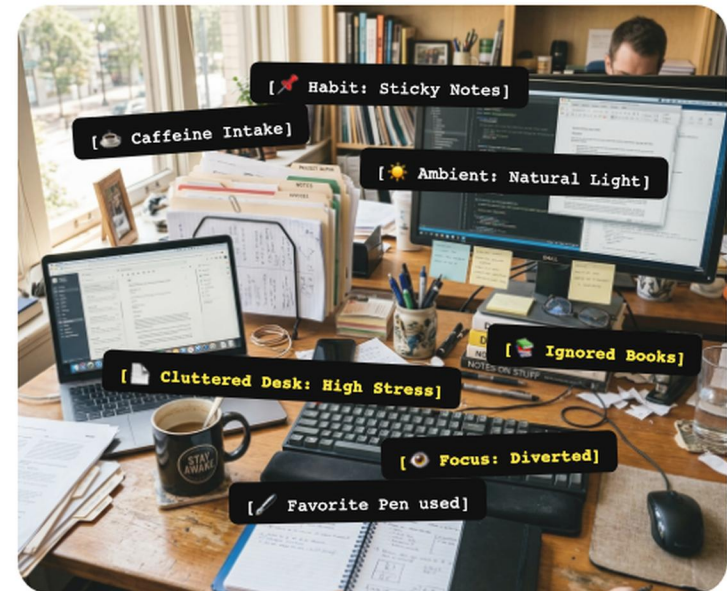
Current AI agents have infinite access to your **digital traces**, but remain completely blind to your **physical reality**. Agents personalize from half the picture.

Digital Dimension



The Gap

Physical World



? Research Questions

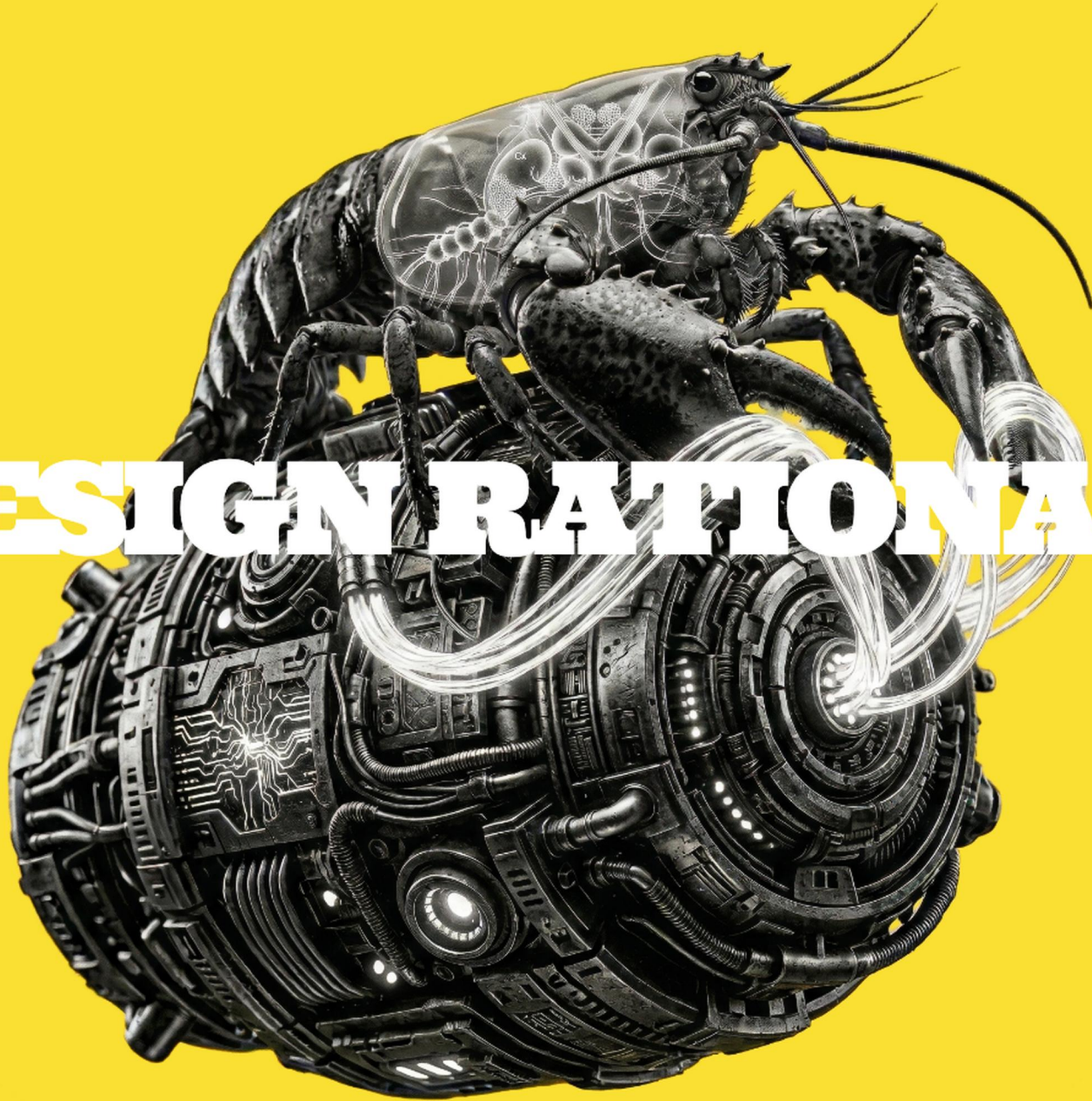
- RQ1** What **physical-world information** becomes **visible** through egocentric perception that remains invisible through digital traces alone?
- RQ2** In what **task & interaction contexts** does this physical-world information meaningfully **improve personalization**, and when does it **introduce noise or redundancy**?
- RQ3** What **design requirements** emerge for integrating **continuous egocentric sensing** into **existing agent memory architectures**?



🔧 Contributions

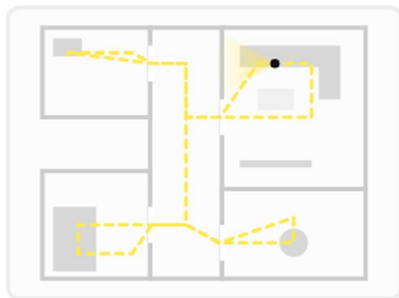
- C1** A two-week **autoethnographic deployment study** examining **what** physical-world information egocentric perception makes visible and **when** it improves agent personalization.
- C2** A **system artifact** integrating egocentric vision into **OpenClaw's markdown-based memory architecture**.
- C3** **Design implications** for physical-world memory integration, covering **storage granularity**, **retrieval visibility**, and **privacy trade-offs**.

DESIGN RATIONALE



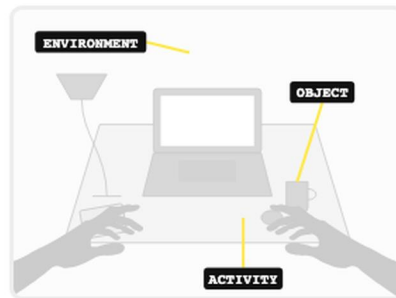
Why 🕶️ Egocentric Vision?

We select egocentric vision—first-person visual capture from a body-worn device—as the **primary** physical-world input, supplemented where useful by audio and inertial data.



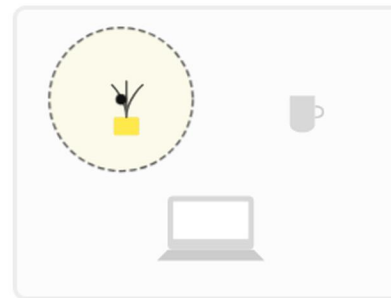
🚶 Mobility

Continuous capture across locations throughout the day, persisting as the user moves between spaces.



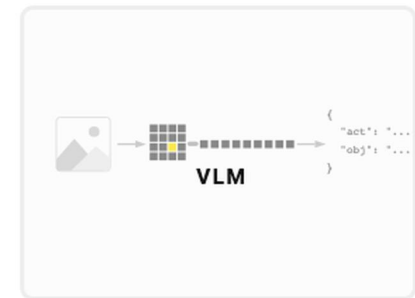
📖 Information Density

A single frame reveals Environment (context), Activity (process), and Objects (entities) simultaneously.



👁️ Implicit Attention

The device's field-of-view serves as a proxy for human attention; in-frame implies active engagement.



🚀 Technical Maturity

Modern VLMs effectively project pixel-level data onto structured semantic logic spaces.

Why 🦞 OpenClaw Integration?

A foundational infrastructure providing a ready-to-use capture ecosystem and a transparent, file-based memory architecture.

📌 Deployed Baseline

Established capture ecosystem with hardware-software parity.

🔓 Open Source

Fully extensible codebase allowing for integration without black boxes.

📝 Markdown Workspace

Human-readable, file-based memory that is inspectable and easy to edit.

WORKSPACE / .MD

- AGENTS.md
- SOUL.md
- USER.md
- memory/
- 2024-11-12.md

2024-11-12 | Digital Self-Log

[10:15] Session Initialization

Device handshake complete. Egocentric stream (v2.1) active. Calibrating depth sensors for SpaceSelfLog workspace.

[10:42] Activity Observation

- User is seated at the main workbench, engaged in hardware debugging.
- Tool identified: Precision screwdriver.
- Target: IMU sensor alignment on the Crustacean-V1 rig.

[11:05] Memory Flush

Critical context saved to **MEMORY.md**. Workspace synced to local git repository.

Inspectable Format
Human-readable text logs that can be audited for privacy.

Easy to Edit & Integrate
Modify logs manually or sync with Obsidian/Notion via git.

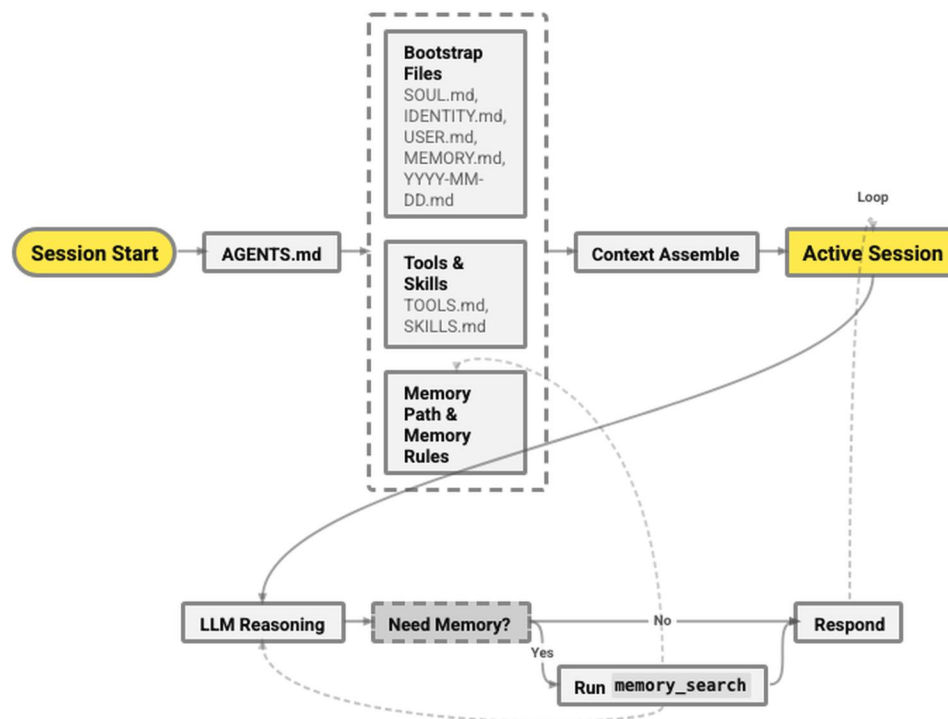
OpenClaw
Steers the model via
structured Markdown system
prompts.

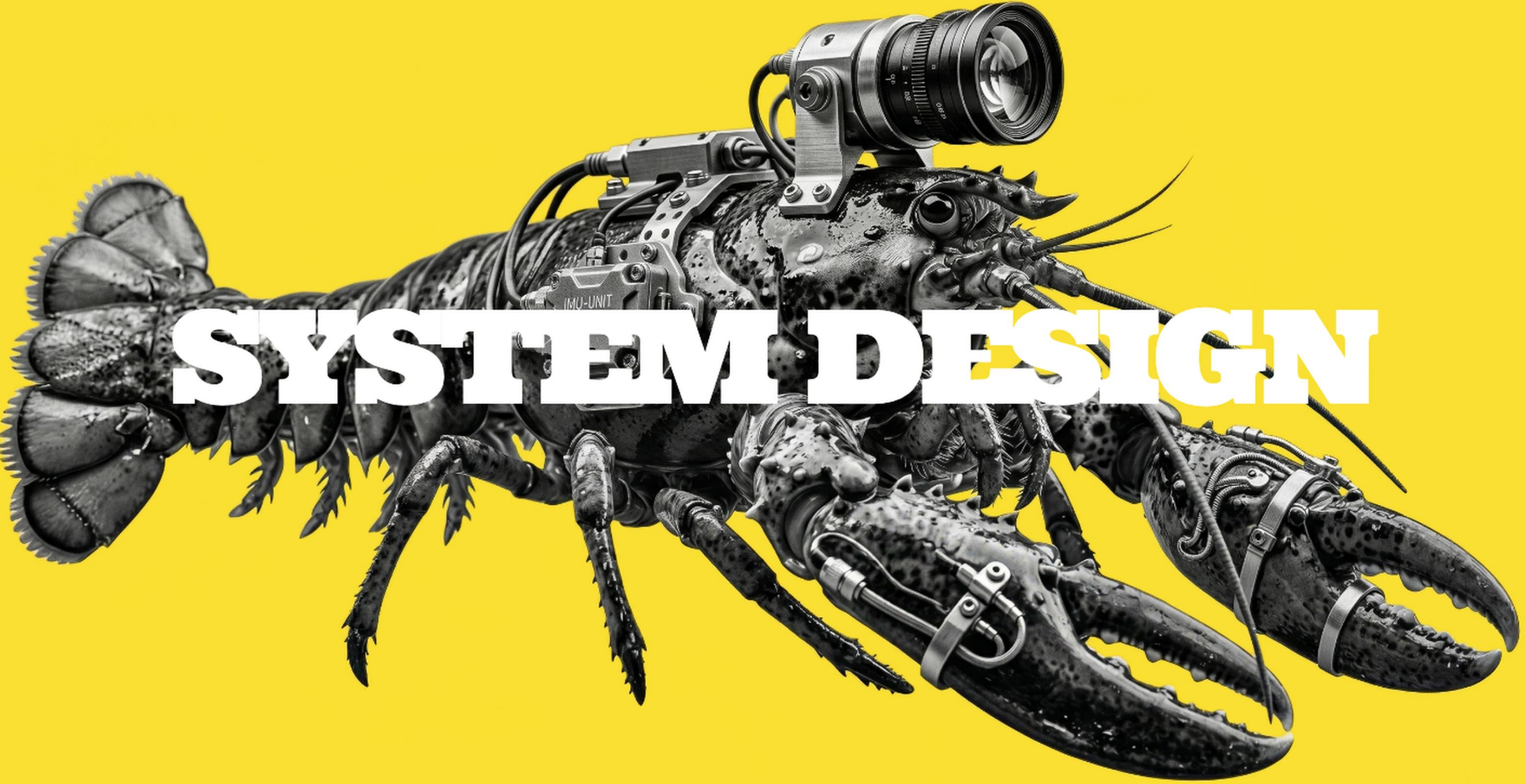


LLM Engine
The underlying model
processing all context.

-  **AGENTS .md** Tool access, behavior rules, response format
-  **SOUL .md** Core persona — tone, values, communication style
-  **IDENTITY .md** Agent name, role definition, relational vibe
-  **USER .md** Known user preferences, habits, background
-  **MEMORY .md** Pinned long-term facts — always in full
-  **YYYY-MM-DD .md** Today + yesterday in full · older logs via vector search

Bootstrap Files:
Agent-maintained — updated
whenever it decides to.
Human-readable & editable
too.



A detailed black and white photograph of a robotic lobster. The lobster is equipped with various sensors and actuators. A large camera lens is mounted on its back, and a smaller sensor unit is labeled "IMU-UNIT". The lobster's claws are also equipped with mechanical components. The entire robot is set against a bright yellow background.

SYSTEM DESIGN



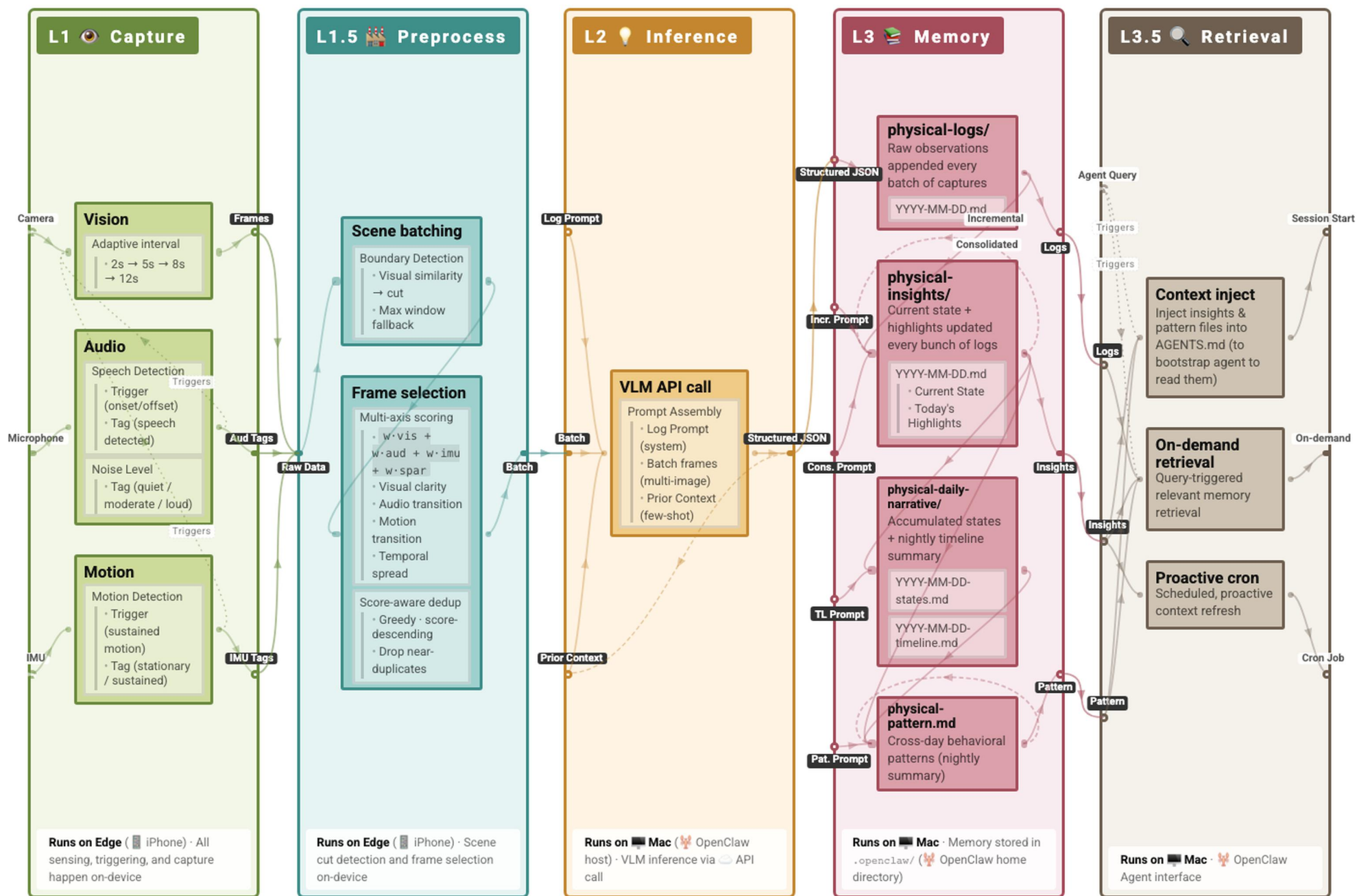
The Core Focus: Strategic Compression

Strategic Pervasiveness: Compressing is the most critical step in this project. Processing every raw frame is prohibitively expensive in power and cost —therefore, **compression is pervasive** throughout the entire pipeline.

Compression = Understanding: To compress is to understand. It requires selection, deconstruction, and reconstruction. Since this involves inherent information loss, the system must **prioritize**. Much like the human brain, which filters infinite sensory inputs to retain only the most meaningful, our pipeline must identify what matters.

Defining Key Events: The essence of compression is the intelligent identification of **Key Events** from the noise. These events are judged via two mechanisms:

- **Autonomous Reasoning:** VLM/LLM-driven judgment guided by specialized system prompts.
- **Human Feedback:** Real-time insights and corrections captured during active chat sessions.



L1 Capture

COLLECTION

COMPRESS

Decides when to capture.

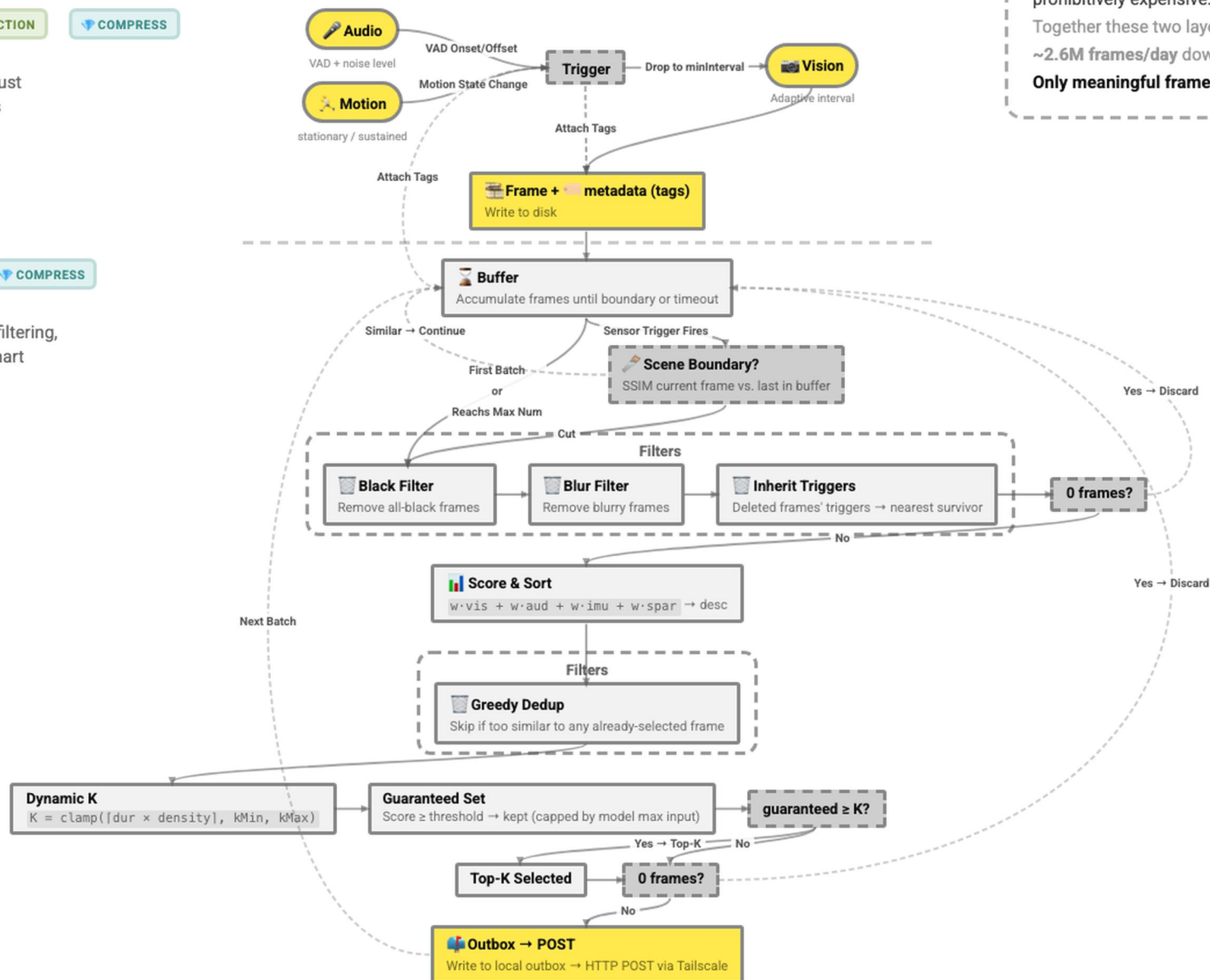
Audio & Motion triggers adjust Vision's interval. All sensors contribute tags.

L1.5 Preprocess

COMPRESS

Decides what to keep.

Scene boundary detection, filtering, importance scoring, and smart frame selection.



Design Rationale

Sending every frame to the cloud is prohibitively expensive.

Together these two layers compress ~2.6M frames/day down to ~300-600.

Only meaningful frames leave the device.

Settings - Parameters

Capture Interval

Min interval ☐

Max interval ☐

Ramp ratio ☐

Audio

VAD sensitivity ☐

Transcription ☐

Noise quiet threshold ☐

Noise loud threshold ☐

IMU

Sustained motion threshold ☐

Variance low threshold ☐

Variance high threshold ☐

Batch / Frame Selection

First batch window ☐

Max window ☐

SSIM boundary threshold ☐

Dedup SSIM threshold ☐

K density ☐

K min / K max ☐

Score threshold ☐

Max output frames ☐

Outbox

Upload endpoint ☒

SpaceSelfLog

Started: 8:05:29 PM Runtime: 03:51:52

SETTINGS

CAPTURE INTERVAL

Min: s

Max: s

Ramp Ratio:

2s 3s 5s 8s 12s

Save

AUDIO

VAD Sensitivity: Low Med High

Transcription: ☒

Noise Quiet Threshold: dB

Noise Loud Threshold: dB

Save

IMU

Sustained Motion Threshold: s

Variance Low Threshold: g

Variance High Threshold: g

Save

Camera: Ultra Wide 0.5x
Angle: 180°
Resolution: High 1080p
Battery: 80% (Charging)

Start Pause Stop Wide 1x Ultra 0.5x Rotate 90° Resolution

Frames 378 Batches 37 Storage 250.6 MB

PIPELINE

Interval: 12s → 12s max

VAD: quiet

Noise: quiet

IMU: stationary

SSIM Debug: 0.721

REFERENCE INCOMING

Buffer: 3 frames

Last Batch: 11:57:20 PM

Real-time Monitor

Sensor Pipeline

Current interval ☒

VAD state ☐

Noise level ☐

IMU state ☐

Frame Processing

SSIM debug ☐

Buffer state ☐

Last batch ☐

Batch Output

Latest batch output ☒

Batch history ☐

Layer 1 latest frame ☐

Device Status

Live camera stream ☐

Battery ☐

Frames captured ☐

Batches sent ☐

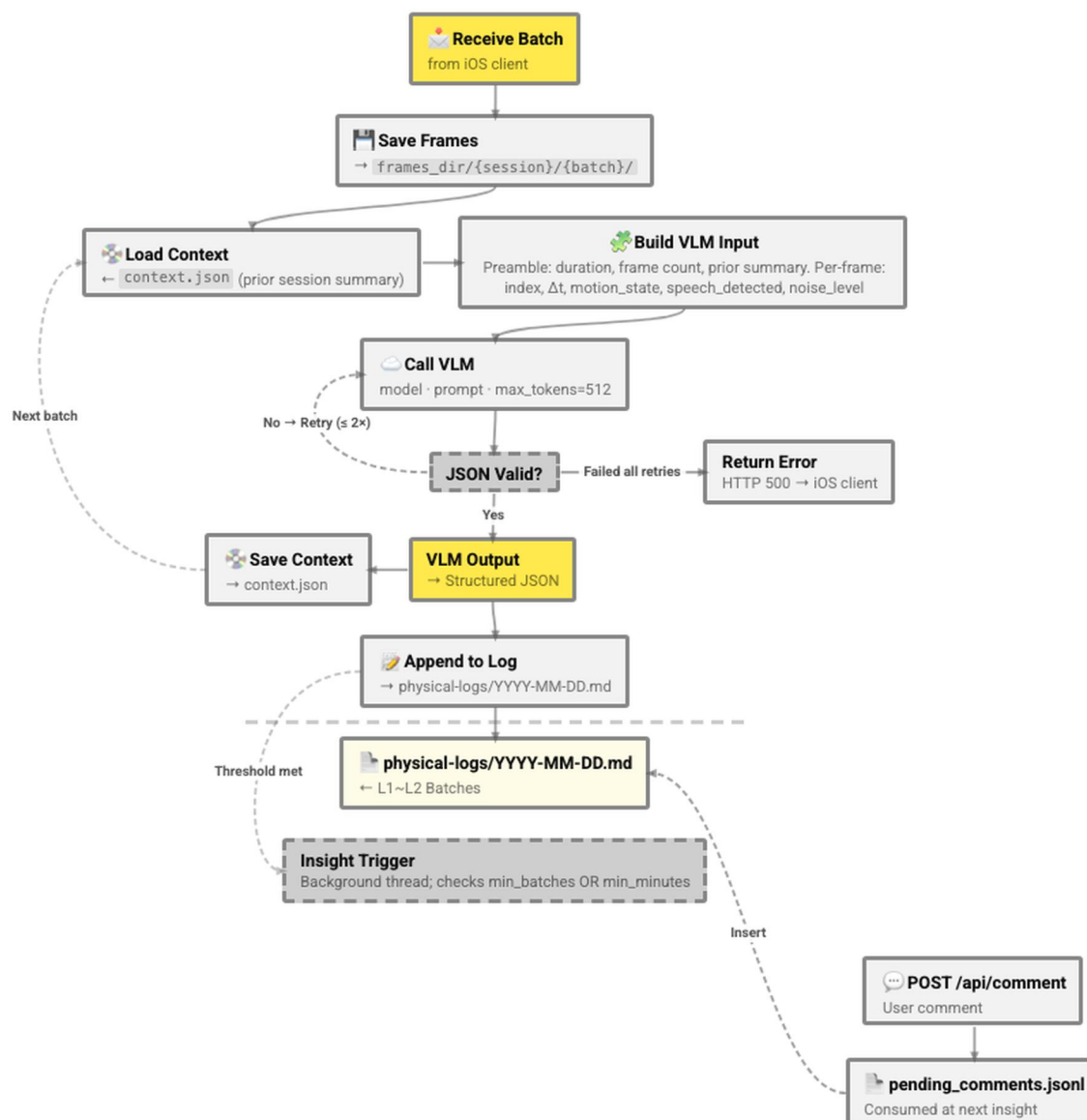
Storage used ☐

L2 Inference

COMPRESS

Per-Batch Analysis

Receives frame batches, runs VLM analysis per batch, maintains session context, writes structured logs.



Design Rationale

- **Context accumulation.** Every VLM call includes batch frames, metadata, and the previous call's result as context.
- **Batch, not stream.** Grouping frames per-scene gives the VLM temporal context to reason about transitions.
- **Insight decoupling.** A single batch rarely carries enough signal for a higher-level summary — that will be handled in the next layer.

L3 Memory

INTEGRATION

OpenClaw Memory Pipeline

Human Feedback

User comments are captured and injected into the insight pipeline at the next trigger.

L3 Memory

INTEGRATION

OpenClaw Memory Pipeline

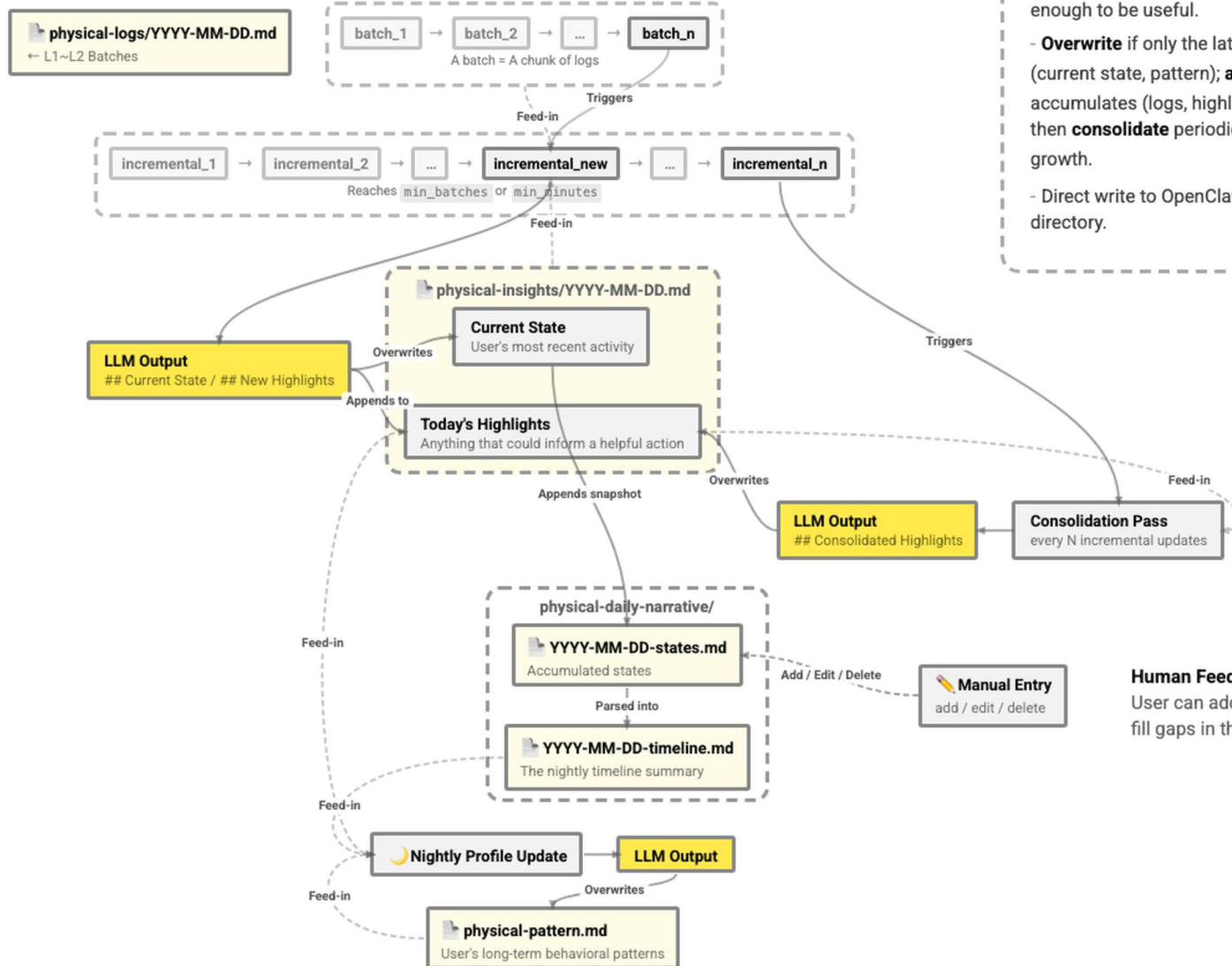
Logs: seconds~minutes snapshots

→ **Insights:** minutes~hours states

& highlights → **Narrative:** daily

timeline → **Patterns:** days~weeks

patterns



Design Rationale

- **4-layer memory.** From short-term logs to long-term patterns.
- Enough context to summarize; timely enough to be useful.
- **Overwrite** if only the latest value matters (current state, pattern); **append** if history accumulates (logs, highlights, states) – then **consolidate** periodically to bound growth.
- Direct write to OpenClaw's memory directory.

Human Feedback

User can add/edit/delete states to fill gaps in the timeline.

Settings - Parameters

Vision Model

Vision provider ☒

Vision API key ☒

Vision model ☒

Text Model (Optional)

Text provider / key / model ☐

Prompts

VLM prompt ☒

Incremental prompt ☒

Consolidation prompt ☒

Pattern prompt ☒

Timeline prompt ☒

Directories

OpenClaw memory/ ☐

Project dir ☐

Frames dir ☐

Scheduling & Auto Insight

Nightly summary time ☒

Min batches to trigger ☒

Min minutes between runs ☒

Consolidation every N ☒

Hook re-inject interval ☐

Alerts

Email + SMTP ☐

SpaceSelfLog Monitor

Iteration Log

Journal

Last batch: 12:53:45 AM

INSIGHT

12:49:54 AM

PATTERN

2026-04-06

CONFIGURATION

Config: /Users/mia/spaceselflog/config.json

Vision Provider

OpenRouter

Vision API Key

Vision Model

mistralai/mistral-small-2603

Test Connection

TEXT MODEL (OPTIONAL)

Text Provider

OpenRouter

Text API Key

(Same as Vision)

Text Model

mistralai/mistral-small-2603

Test Text Connection

Save Provider Settings

VLM Prompt (per-batch)

You are an egocentric perception system analyzing first-person

Batch Feed

INSIGHT Today 4 incremental 0x consolidated Next consolidation in 6/10 Batch 5/10

Time 4/30 min 5 batches - 26 min

12:53:45 AM 2026-04-10T04-53-26Z ok mistral-small-2603 3.1s 1/1 frames - 2026-04-10_0

```
{ "activity": "screens-only monitoring with new manual interaction", "location": "home office workspace", "objects": [ "two 24-inch monitors displaying matrix-style diagnostics and dashboard overlays", "mechanical keyboard in view but unused", "small rubber duck on left workspace edge near stack of papers", "red sports drink bottle capped and slightly repositioned on right edge", "green mesh decanter and water pitcher present but untouched", "mug of tea steaming lightly", "folders and manuals stacked behind monitor" ], "social_context": "alone; no one else in foreground or visible in peripheral", "notable_events": [ "user briefly touched the red sports drink bottle, reorienting its angle toward the decanter; no sustained input on keyboard or touchpad detected", "observation": "The user remained seated and stationary during the 12-second interval, visually aligned with the two 24-inch diagnostic monitors as they continued to render matrix-style outputs and dashboard views. The hands were not seen over the keyboard or mouse, reinforcing the prior pattern of passive screen observation rather than active input. A small rubber duck on the left edge lay unchanged, suggesting its role as a fidget or visual marker rather than interactable tool. The red sports drink bottle was capped and had been rotated roughly 30 degrees from its prior orientation, implying a deliberate but momentary adjustment of a hydration habit that usually places the bottle beside the keyboard center. Steam rising from the mug indicated a freshly poured hot beverage, yet the user did not reach for it, reinforcing the inactivity on peripherals. This brief event underscores a quiet, deliberate monitoring phase where ergonomic tweaks—like bottle position—are enacted without interruption to the primary task of system observation." ] }
```

11:47:19 PM 2026-04-10T03-46-54Z ok mistral-small-2603 3.2s 1/2 frames - 2026-04-09_2

Daily Log

Insights

States

Timeline

Pattern

Events

2026-04-10

CURRENT STATE (AS OF 00:49)

The user is holding a steady, late-night diagnostic session in the home office, still seated at a dual-monitor workstation with both terminal output and a server-rack diagram on screen. They remain alone, intermittently hydrating with a water bottle while the now-repositioned Red Bull energy drink sits unopened in immediate reach. Minimal environmental disruption—primarily the presence of a cat in the periphery—has not altered the user's high-focus pattern.

TODAY'S HIGHLIGHTS

- Sustained terminal command-line interaction with visible cursor activity (none)
- Zero-calorie Coca-Cola Zero Sugar can placed at the edge of work surface as a recurring source of caffeine during high-focus technical work (none)
- Ceramic mug with light brown contents and spoon maintained as a hydration artifact through multiple debug cycles (none)
- Transition from passive error log review to active terminal synthesis requiring both keyboard and mouse input, indicating a shift from analysis to hands-on debugging (none)
- Open hard-shell rolling suitcase initialized on floor adjacent to desk, with stacked grey clothing items being refolded during packing process while technical displays remain live in the background
- White ceramic mug containing hot beverage introduced to workspace after luggage packing interruption, placed beside the keyboard
- Workspace clutter expanded to include scattered drink containers (empty disposable cup sleeve, lid, base, and iced beverage cup with purple straw) alongside nutritional guides (e.g., Algran 150) during marathon technical session
- Remote meeting/webinar interface visible on laptop screen

Real-time Monitor

Batch Feed

☒ Incoming batches

☒ VLM latency

☐ Insight trigger bar

Log Files

☒ Daily log

☒ Insights

☒ Pattern

User Annotations

☐ States

☐ Timeline

Event Log

☒ Batch events

☒ Insight events

☒ Pattern events

☐ Errors

L3.5 Retrieval

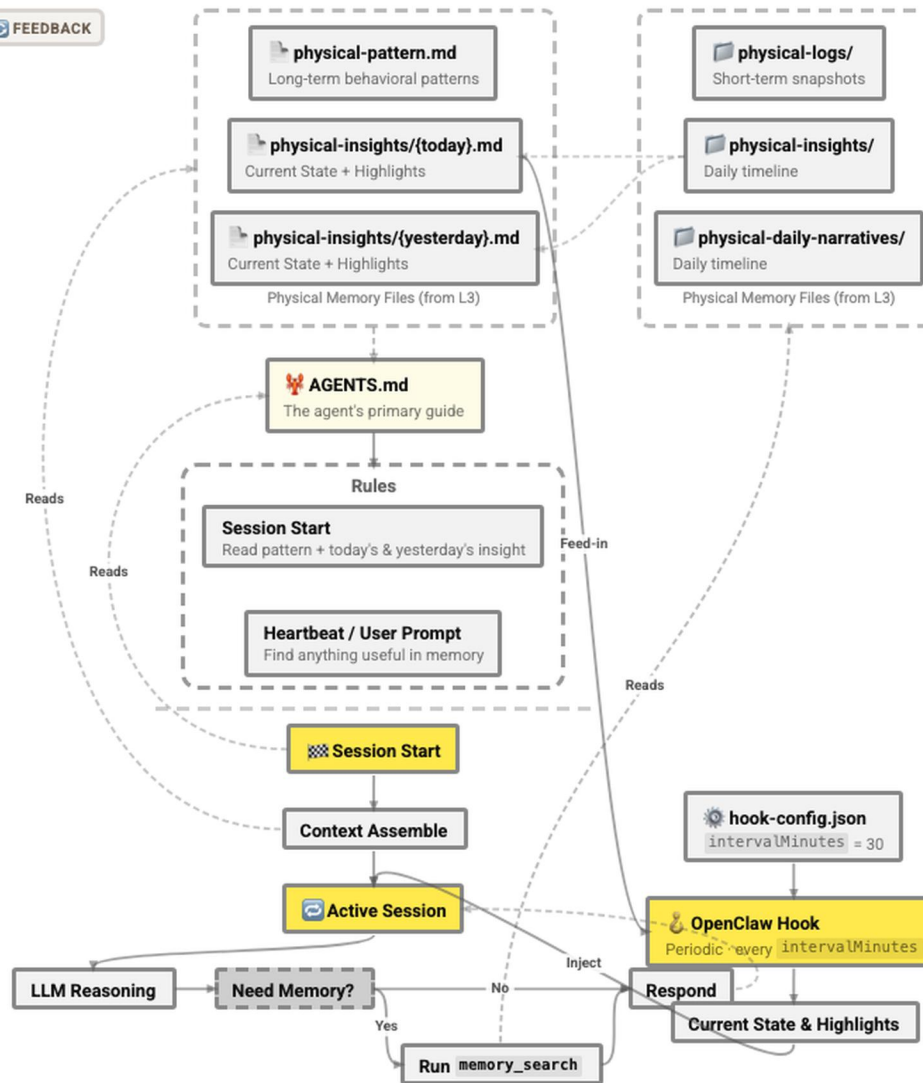
INTEGRATION

FEEDBACK

When the agent uses memory

& How the agent accesses memory

Declarative rules tell the agent when and what to read from memory files.



Design Rationale

- **Utilize OpenClaw's built-in memory system.** AGENTS.md rules let the agent pull memory at session start; the proactive hook pushes fresh context mid-session without user re-prompting.
- **Cron keeps context fresh.** Long sessions drift — periodic injection ensures the agent always sees the latest physical state.

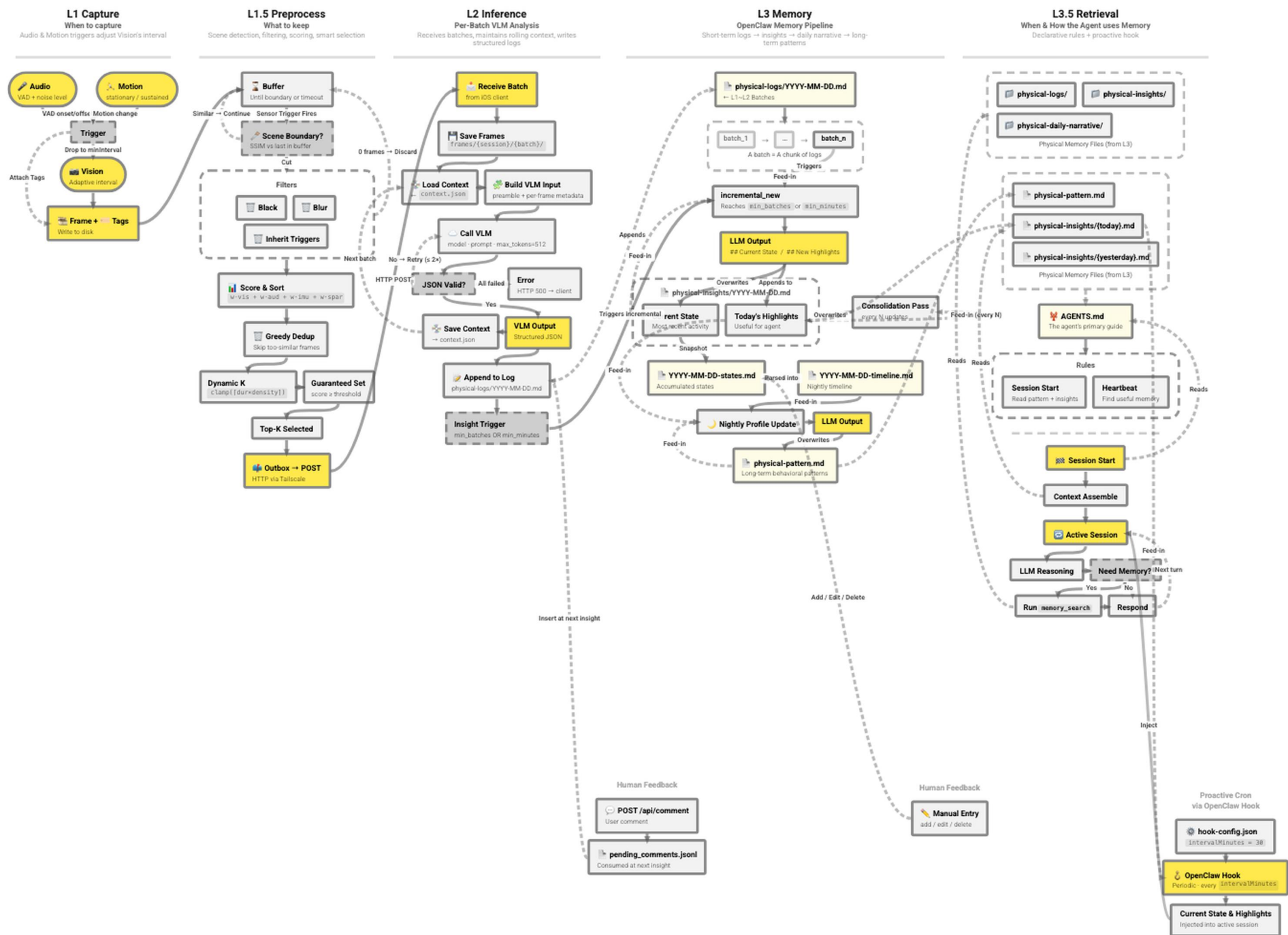
OpenClaw Agent

Where the memory is used

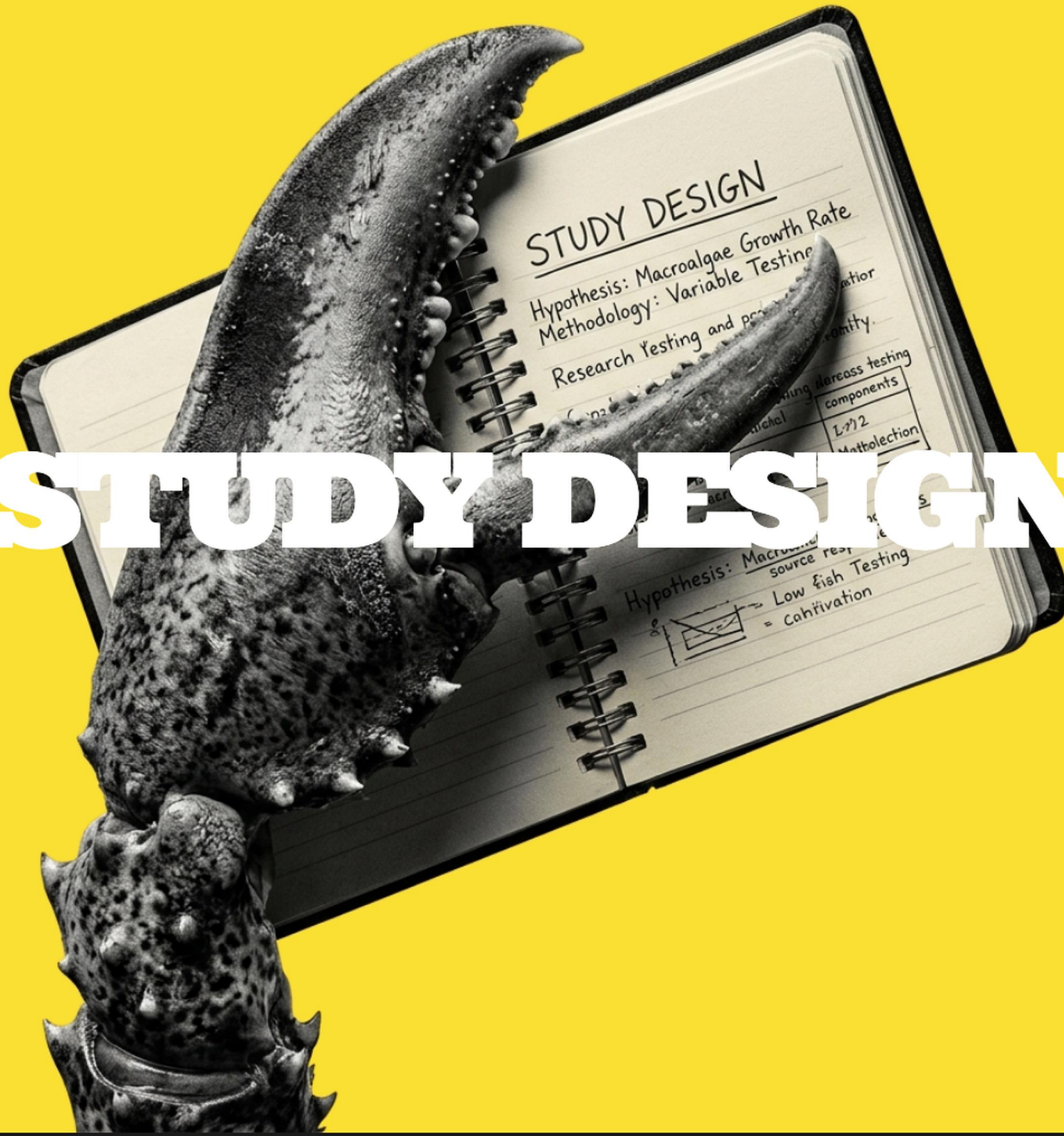
Proactive Cron

Via OpenClaw Hook

Background hook pushes fresh insight into the active session on a schedule.



STUDY DESIGN



Study Design

A two-phase framework supporting exploratory iteration on design variables and rigorous observation of agent behavior.

Phase 0: Calibration

APPROX. 1 WEEK · ACTIVE ITERATION



System Stabilization

Active tuning of over a dozen design variables spanning capture timings, pre-process thresholds, and prompt structures.



Design Iteration Log

Documenting modifications, rationales, and before/after states to construct the primary dataset for emerging design implications.

Phase 1: Observation

2 WEEKS · FROZEN PARAMETERS



Passive Observation

Using the agent organically for daily tasks to trace whether physical-world contexts surface without deliberate steering.



Structured Probes

Deploying 1–2 daily requests crafted around captured logs to evaluate agent utilization of physical traces when explicitly relevant.



Proactive Cron

Tracking the agent's autonomous judgment to autonomously trigger contact or deliberately decline acting on new physical-world activity updates.

Data Collection

A combined methodology integrating system **telemetry** with researcher-produced **autoethnographic** reflections.

Automated Data Sources

Perception Log

Phase 1

The complete record of unfiltered VLM outputs. Represents everything the egocentric capture pipeline made visible.

→ Feeds RQ1 (Content Classification)

4-Tier Memory Output

Phase 1

Records what OpenClaw gracefully retained, distilled, or discarded across physical-logs, physical-insights, and physical-pattern.

→ Feeds RQ3 (Memory Flow Analysis)

Interaction Transcripts

Phase 1

Continuous transcript corpus logging passive use, structured probes, and autonomous cron behaviors.

→ Feeds RQ2 (Interaction-Level Coding)

Researcher-Produced

Design Iteration Log

Phase 0

Documents every permissive pipeline modification alongside its rationale and before/after output comparison.

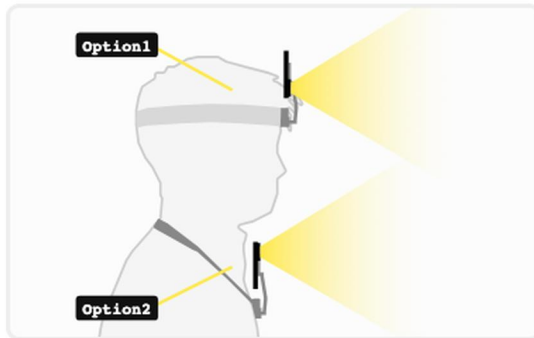
→ Feeds RQ3 (Thematic Analysis)

Autoethnographic Journal

Phase 0 · Phase 1

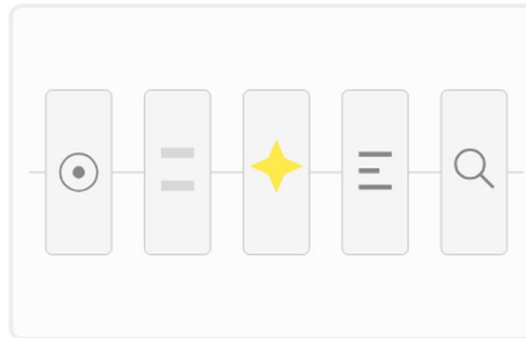
Twice-daily structured entries (mid-day review and end-of-day reflection) capturing interpretative judgments that automated logs cannot represent. Covers interaction assessments, missed opportunities, noise, and design reflection.

→ Feeds All RQs (Interpretive Anchor)



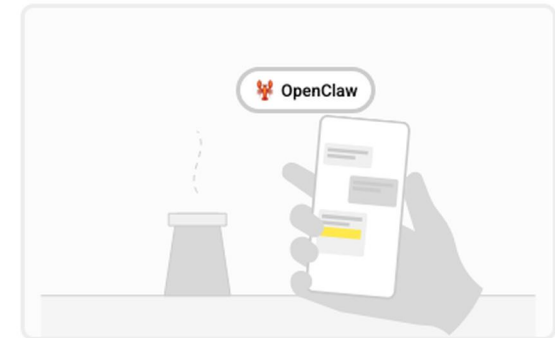
1. Continuous Capture

Wearing the egocentric device actively throughout the day, providing an "always-on" base log of physical reality.



2. System Pipeline

Transforming continuous egocentric traces through 5 uniformly architected data layers (L1 through L3.5) into deep, interactive OpenClaw knowledge assets.



3. Passive & Structured Usage

Using the agent organically for physical-world memory surfaces, while reviewing raw logs to explicitly craft loaded prompts and test contextual awareness.



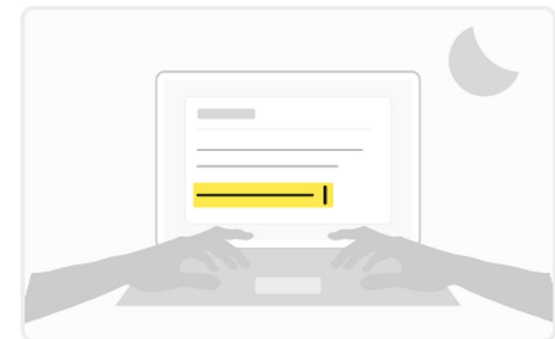
4. Proactive Encounters

Experiencing system-triggered interruptions during unrelated physical actions, evaluating unprompted judgment.



5. Mid-Day Assessment

A rapid 5-minute interaction scoring session to capture interpretative judgments while memory is completely fresh.



6. End-of-day Reflection

A deep 15-minute alignment comparing the logged "system reality" against the nuanced "lived context".

#1 Why an Off-the-Shelf Smartphone?

It has everything required natively: a high-quality camera, microphone, IMU, Wi-Fi, and reliable battery life. Developing a native iOS Swift application offers significant agility compared to building custom hardware from scratch or hacking closed smart-glasses ecosystems.



#2 Why Chest-Mounted?

Ultimately opted for the chest-mounted **Option 2** over the head-mounted ~~Option 1~~. While sacrificing some attentional cues from neck rotation, it provides **Greater Stability** and **Enhanced Comfort**. This allowed me to forget the device's presence completely and behave entirely naturally during the study.

Iteration Form — Inputs

Core Variables

Component Changed ●

Custom Variable ●

Narrative Context

Change Description ●

Rationale ●

Behavioral Shift

Before Output ●

After Output ●

Design Iteration Log

Phase 0 calibration record

[← Monitor](#)

NEW ENTRY

Date

05/01/2026

Variable / component changed

— select —

Change description *

What exactly was changed? (values, logic, wording...)

Rationale *

Why was this change made? What problem did it address?

Before

After

Output / behavior before the change

Output / behavior after (or expected)

Save entry

Clear

HISTORY 4

maxInterval

Apr 1, 07:59 PM

×

CHANGE

max capture interval from 16 -> 12; min capture interval from 6 -> 2

RATIONALE

I found that min_interval must be sufficiently small, and the interval of the initial ramp step must also be sufficiently small to ensure fast response and continuous capture after triggering. Consequently, max_interval must also be adjusted correspondingly.

BEFORE

Occasionally, rapid continuous capture is not triggered immediately following scene changes or action triggers.

AFTER

—

rampRatio

Apr 1, 07:59 PM

×

CHANGE

2 -> 1.6

RATIONALE

Also to ensure the initial ramp steps are small enough for fast response and continuous capture after triggering.

Audio VAD Sensitivity

Apr 1, 07:59 PM

×

CHANGE

Low -> Med -> High

RATIONALE

I tested several different volume levels. High sensitivity is required to robustly trigger while speaking.

BEFORE

Unable to detect speech; threshold is never exceeded.

AFTER

—

Audio Noise Threshold

Apr 1, 01:08 AM

×

CHANGE

Context & History

Historical Reference

Entry History ●

Before / After Traces ●

Component Tracking

Timing & Intervals ●

Frame Selection ●

Prompt Engineering ●

Journal Form — Inputs

Assessment Type

Mid-day Check-in

End-of-day Reflection

Structured Fields

Interaction

Assessments

Utilization Summary

Missed Opportunities

Technical Narrative

Pipeline Behavior

Phase Mod Log

Noise / Irrelevance

Autoethnographic JournalPhase 0 + Phase 1

← Monitor

Mid-day Check-in-5 min

End-of-day Reflection-15 min

Date8/11/2026PhasePhase 1 Week 2 — proactive hr

INTERACTION ASSESSMENTS

INTERACTION 1

Brief description of this interaction

Physical context surfaced?Was it useful?

Reasoning / what should have surfaced / why it helped or didn't...

+ Add interaction

STRUCTURED PROBE INTENT

Why a perception log entry was selected and what the probe was designed to test (leave blank if no probes today)

e.g. Perception log showed researcher worked from café three afternoons. Probe: 'suggest a workspace for this afternoon' — testing whether café pattern surfaces unprompted.

Save entry

HistoryToday's Conversation

AllGroupDM

1 msg6 DMs · updated 02:54 PMRefresh

cron · 02:32 PM

New session · 02:32 PM

model: glm-6 · 02:32 PM

thinking: low · 02:32 PM

[custom] 02:32 PM

(cron:ccd1e9b4-e2e8-40ad-9246-63a0edc95432 Physical Insight Check) Read memory/physical-insights/YYYY-MM-DD.md for today's date (use the current date). If the file doesn't exist, stop and do nothing. If it exists, analyze the latest insights for anything timely or actionable that Chris should know about right now — e.g. unusual activity, health signals, overdue breaks, concerning patterns. If nothing warrants interrupting him, reply with just 'NO_REPLY'. If something is genuinely useful/urgent, send a concise message to Chris via this session (it will route to his Telegram DM). Keep it brief and casual. Don't force a message if there's nothing worth saying.

Current time: Saturday, April 11th, 2026 — 2:32 PM (America/New_York) / 2026-04-11 18:32 UTC

thinking... (click to expand) · 02:33 PM

read /Users/mia/openclaw/workspace/memory/physical-insights/2026-04-11.md · 02:33 PM

read · 02:33 PM

thinking... (click to expand) · 02:33 PM

This is a lot of detail. Let me look for anything actionable or concerning:

- It's Saturday 2:32 PM

- He's been in sustained computing sessions with minimal breaks

- Comfort food snacking (Pop Tarts, Pringles) throughout

- He did move to a yoga mat briefly for physical cool-down

- Generally seems like a long deep-focus day but with some breaks

Nothing genuinely urgent or concerning. The yoga mat break shows self-regulation. Nothing unusual enough to interrupt him.

02:33 PM

NO_REPLY

Context & History

Reflection Support

Today's Conversation

Filter: All / Group / DM

Tool Result Trace

Historical Reference

Entry History

Longitudinal Sync

Agentic Metadata

Thinking Blocks

Hook Injection Trace

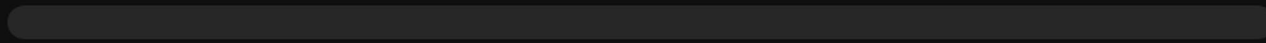
34



RESULTS

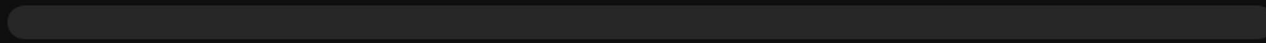


Processing Space-Time Data 0 / ...



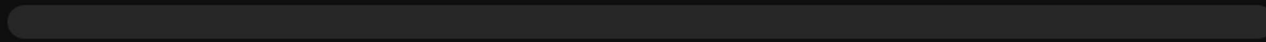


Processing Space-Time Data 0 / ...



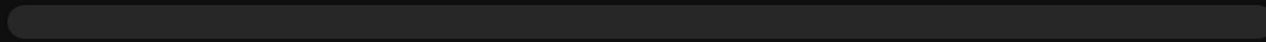


Processing Space-Time Data 0 / ...





Processing Space-Time Data 0 / ...





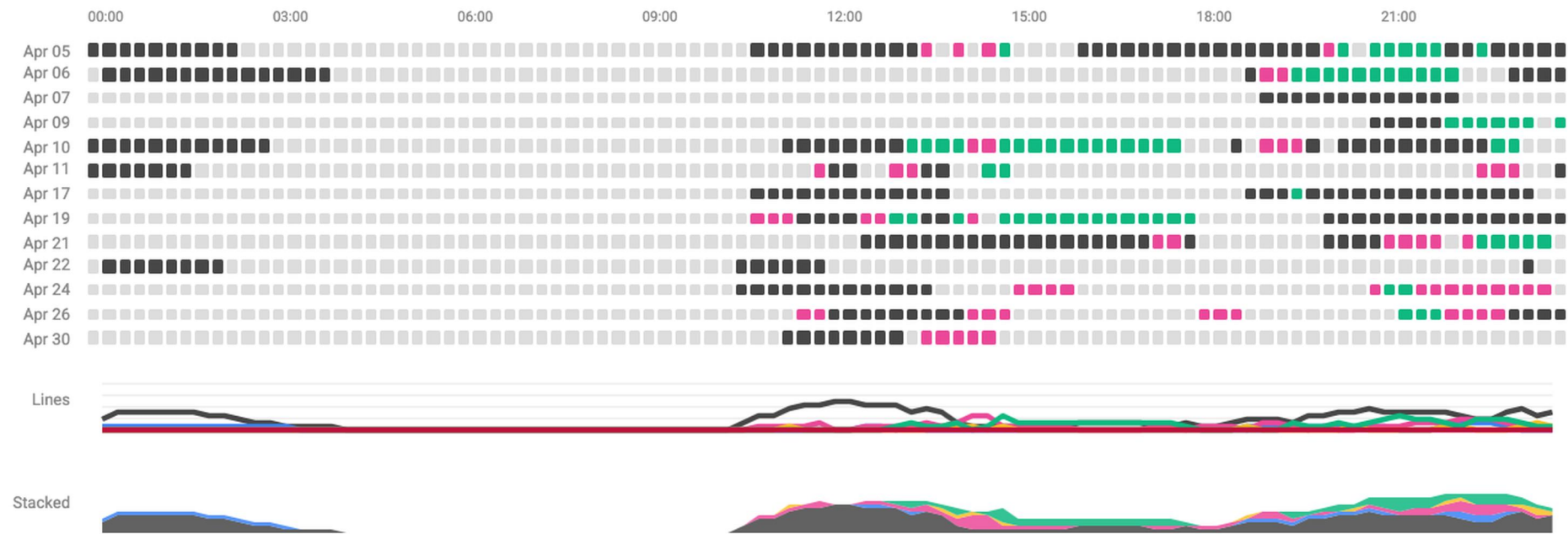
Processing Space-Time Data 0 / ...



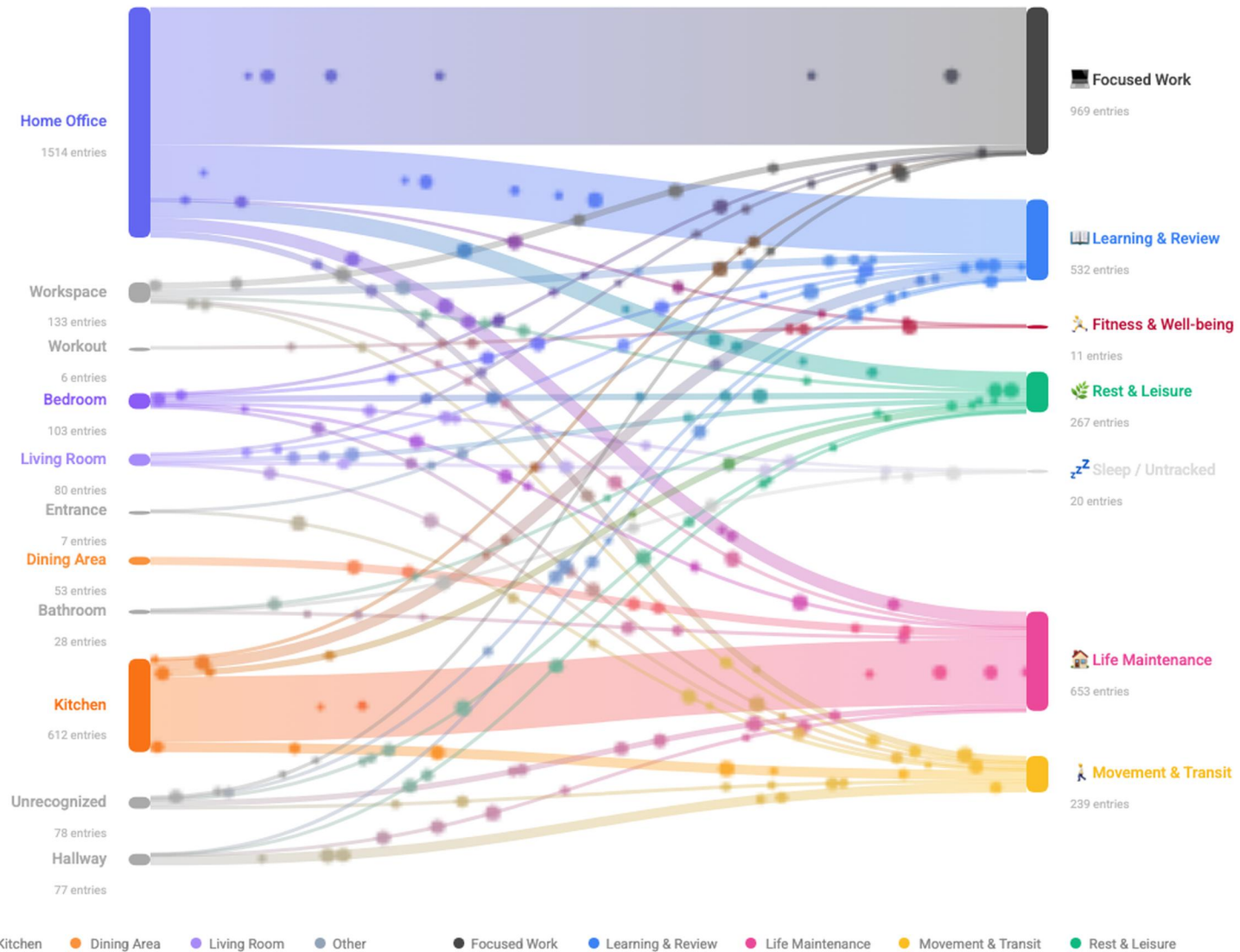


Processing Space-Time Data 0 / ...





15-minute intervals across six standardized cognitive states — aggregated across all days.



Each band shows how time logged in a location flows into cognitive activities. Band width = number of log entries.



Physical Rhythm Map: Sidebars visualize utilization ranking and hourly activity load for each zone.

Public Areas

Dining Area

57



Life Maintenance 93%
Learning & Review 4%
Movement & Transit 4%

Kitchen

613



Life Maintenance 70%
Learning & Review 11%
Movement & Transit 11%

Living Room

84



Rest & Leisure 40%
Learning & Review 25%
Movement & Transit 23%

Entrance

9



Movement & Transit 44%
Learning & Review 33%
Fitness & Well-being 22%

Hallway

77



Movement & Transit 84%
Learning & Review 8%
Life Maintenance 4%

Unrecognized

80



Life Maintenance 49%
Movement & Transit 19%
Learning & Review 16%

Private & Work

Home Office

1515



Focused Work 60%
Learning & Review 24%
Rest & Leisure 8%

Bedroom

103



Rest & Leisure 41%
Movement & Transit 21%
Sleep / Untracked 13%

Workout

6



Fitness & Well-being 100%

Workspace

133



Learning & Review 38%
Focused Work 32%
Life Maintenance 14%

Bathroom

32



Life Maintenance 59%
Rest & Leisure 16%
Sleep / Untracked 13%



● Focused Work
 ● Learning & Review
 ● Life Maintenance
 ● Movement & Transit
 ● Rest & Leisure
 ● Sleep / Untracked
 ● Fitness & Well-being
 ● ? Other

Integrated Activity Map: Room glow indicates dominant behavior. Sidebars show high-resolution breakdown per zone.

What the Agent Learned About Me

Extracting language-based understanding from continuous spatial and physical patterns.

Source: memory/physical-pattern.md

AGENT INSIGHT

"Beverage rotation synced to workload: Espresso (design/app pivot) → soda/orange cream (deep focus) → tea (log review) → latte (image browsing)."

SENSORY PATTERN

Observed in 40 of 44 late sessions

AGENT INSIGHT

"Late-night kitchen looping consolidated: Desk ↔ kitchen ↔ snacking ↔ beverage ↔ return."

SPATIAL LOGISTICS

Observed in 41 of 44 late cycles

AGENT INSIGHT

"Paper-digital bridge sustained: Printed maps clipped to desk, sticky-note archiving, OCR/scan placement, colored sticky pads, hospital meal tray protocols."

WORKFLOW BRIDGING

Observed in 49 of 53 recent workdays

AGENT INSIGHT

"Persistent delayed sleep-wake cycle: Rest ~01:00–01:30, wake ~11:20."

RHYTHM DYNAMICS

Observed 57 of 58 days

AGENT INSIGHT

"Desk periphery expansion: Dual-laptop desk → reclined bedroom → kitchen → hybrid studio integration."

WORKSPACE EXPANSION

Observed 22 of 23 late nights

AGENT INSIGHT

"Analog-digital harmonization daily: Green notebook present during snacking; sticky-note task markers on refrigerator; dual-laptop editing of Physical-World Observation document."

ANALOG-DIGITAL HARMONIZATION

Observed in 46 of 46 recent evenings

RQ1

VISIBILITY &
BLIND SPOTS

FINDING

Physical capture reveals three invisible dimensions: 🏠 **spatial semantics** (desk vs. sofa context), ☕ **subconscious object rituals** (coffee prepared then forgotten), and ⌚ **in-between states** — the drifts between tasks that no digital log ever records.

LIMIT

🔄 Rolling memory handles statistics, not the **why**. True insight requires a dedicated 🧠 **reflection & deduction module**: structured prompting, data visualization, and full user modeling. An LLM cannot self-generate wisdom by questioning itself alone.

RQ2

IMPROVEMENT
VS. NOISE

PEAK

The agent inferred an 🧑 **unannounced offline activity** (rock climbing) purely from prior patterns. It also sent 🍽️ meal reminders based on learned rhythms. Physical memory enables ✨ **contextual inference** beyond what is explicitly shared.

NOISE

🤖 VLM hallucinations are pervasive (e.g., "cleaning fridge coils"). Proactive messaging compounds this — without 💬 **interaction memory**, the agent repeats itself. Ideal design: track history, detect non-responses, and follow up only once.

RQ3

EMERGING
DESIGN RULES

REQ 1 & 2

📧 **Capture redundantly; filter automatically.** Human review is impossible — feedback must be interaction-based. The system over-relies on the 🔍 **Insights layer**. A 🔗 **traceable inference chain** is essential for correcting the whole pipeline.

REQ 3 & 4

Friction is relational. Overhead is accepted if the user sees the agent as a 🧡 **companion** rather than a tool. 🚫 **Third-party privacy** remains a non-negotiable requirement, demanding a physical kill switch for sensitive contexts.

Moments of Connection & Disconnection

Extracting two defining case studies from the two-week autoethnography.



A Moment of Connection

THE SCENARIO

A prolonged gap in the physical recording logs during the afternoon.

AGENT'S REACTION

Instead of failing or outputting empty data, the agent cross-referenced past dialogue and spatial habits, proactively asking:

"Did you go rock climbing?"

SIGNIFICANCE

This marked the exact threshold where the system transitioned from a passive bystander diary into an empathetic, context-aware companion.



A Moment of Disconnection

THE SCENARIO

A standard, cluttered daily working environment at the desk.

AGENT'S REACTION

The Vision-Language Model (VLM) fixated on random background objects and screen glare, confidently concluding:

"You are cleaning the refrigerator coils."

SIGNIFICANCE

A stark reminder of the "over-interpretation" flaw in current VLMs: the tendency to forcibly impose deep meaning onto meaningless physical clutter.

System Limitations

Bottlenecks identified through the two-week autoethnographic deployment.

01

Form Factor & The Hawthorne Effect

Head-mounting induced constant awareness; chest-mounting mitigated this, but hardware presence remains a barrier.

02

Third-Party Social Boundaries

Personal privacy is unproblematic, but filming others (e.g., video calls) creates severe social friction requiring manual intervention.

03

The Traceability Problem

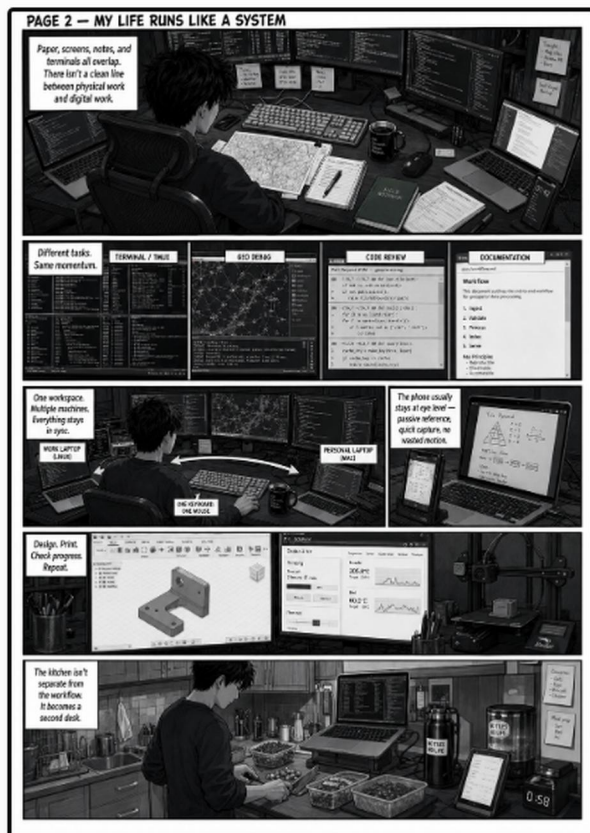
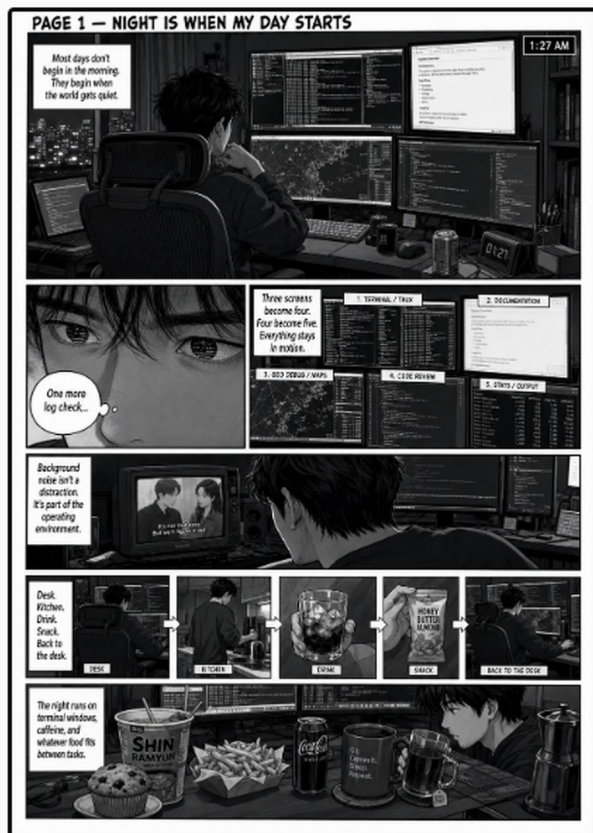
Rolling memory compression inevitably destroys raw details. Tracing a high-level insight back to its origin is currently impossible.

04

VLM Over-Interpretation

Current vision models struggle with passive objects, forcibly imposing profound narratives onto meaningless physical clutter.

The Biographer Has Opened Its Eyes.



APPENDIX

Behavioral Profile

Circadian & Focus Rhythm

- **Persistent delayed sleep-wake cycle:** Rest ~01:00–01:30, wake ~11:20 — observed 57 of 58 days.
- **Deep-focus stillness bursts >12s:** Terminal multiplexing, terminal-log review, CAD/3D-print workflows, medical imaging review — observed 59 of 59 days.
- **Late-night kitchen multispace navigation:** Workstation → kitchen → café → living space — observed 26 of 28 late sessions.
- **Desk periphery expansion:** Dual-laptop desk → reclined bedroom → kitchen → hybrid studio integration — observed 22 of 23 late nights.
- **Hybrid desk transplantation:** Living-room desk sustains tech work, meal prep, video calls, docs — observed 27 of 27 post-travel sessions.
- **Post-lunch focus drop-off:** Secondary workspace routines post-13:00 tied to kitchen/transition lulls — observed 6 of 8 days.

Core Workflow Patterns

- **3+ display superblocks:** Terminal multiplexing + Kubernetes logs + geospatial debugging + documentation + medical imaging review — sustained across 58 of 59 days.
- **Media-work adjacency reinforced:** Korean/English streams on secondary CRTs during deep sessions — observed 55 of 59 days.
- **Finance/authoring pivot anchored:** Late-night document composition with lyric-style breaks; banking during breaks — observed 20 of 20 post-travel evenings.
- **3–5 active displays standard:** Sustained across 57 of 59 days.
- **Paper-digital bridge sustained:** Printed maps clipped to desk, sticky-note archiving, OCR/scan placement, colored sticky pads, hospital meal tray protocols — observed 49 of 53 recent workdays.
- **Domain-shift automation intact:** Debugging → Kubernetes logs → geospatial pipeline oversight → medical imaging review — observed 46 of 53 sessions.
- **Dual-monitor data coupling:** Statistics output panel paired with code review/diffs during sustained editing spans — observed 19 of 21 late cycles.
- **Single-laptop pivot routines:** Emerging post-12:30 transitions to laptop-centric workflows (finance, benchmarks) — observed 15 of 18 afternoons.

Hardware & Digital Ecosystem

- **Terminal multiplexer dominance:** Cross-display iTerm2/WezTerm with active mouse pointers — observed 56 of 59 days (stable).
- **Multi-laptop orchestration:** CharlieOS foreground ↔ macOS/CharcoalOS background; keyboard/mouse swap across hardware — observed 27 of 27 recent sessions.
- **Input modality discipline:** Vertical phone stand at eye level for passive reference; canvas tablet remained disconnected — observed 25 of 26 idle/review phases.
- **Smartphone to laptop mirroring:** Screen duplication during photo/camera workflows, MIDI controller integration — observed 17 of 17 creative sessions.
- **OctoPrint monitoring loop:** Fusion 360 CAD ↔ printer temps ↔ progress tracking — observed 13 of 14 recent nights.
- **USB headset engagement:** Audio-based research/note-taking with peripheral annotation — observed 7 of 8 late sessions.

Spatial & Domestic Logistics

- **Display density adjusted for domestic terrain:** 3–4 active displays typical; workspace perimeter adapted for childcare, grooming, and renovation collateral — observed 56 of 58 days.
- **Kitchen-secondary workspace cemented:** Meal-batching rituals documented by phone with active terminal sessions — observed 38 of 43 domestic days.
- **Renovation leak-through still active:** Kitchen cabinet frame inspection, duct alignment, cavity verification — observed 26 of 27 late sessions.
- **Rolling suitcase mobile periphery:** Partial unloads; kitchen displacement routine — observed 30 of 31 evenings.
- **Printer bedside presence:** Stationary design/print output within arm's reach of primary desk — observed 25 of 26 recent sessions.
- **Secondary workspace integration:** Post-14:00 shifts to desk + CRT + paperwork staging areas — observed 7 of 8 recent transitions.

Sensory & Sustenance Patterns

- **Late-night diet progression reinforced:** Muffin → rice/burger → ramen → structured meals → fries → baking snacks → hospital meal assembly — observed 37 of 38 late days.
- **Beverage rotation synced to workload:** Espresso (design/app pivot) → soda/orange cream (deep focus) → tea (log review) → latte (image browsing) — observed 40 of 44 late sessions.
- **Smart snacking stream validated:** CRT-adjacent bento → kitchen-island bento → three-monitor desk bento with hydroponic bottles — observed 42 of 44 late cycles.
- **Late-night kitchen looping consolidated:** Desk ↔ kitchen ↔ snacking ↔ beverage ↔ return — observed 41 of 44 late cycles.
- **Caffeine-creativity coupling intact:** Latte before design/app pivot; macro pad + numeric keypad during modeling/finance — observed 25 of 27 recent cases.
- **Structured hydration rituals:** Clancy's-branded drinkware rotation during desk meals — observed 14 of 15 analytical blocks.
- **Post-lunch carb coupling:** Shift to tortilla chips/crackers during afternoon financial modeling — observed 11 of 15 sessions.
- **Nutrient dosing timing:** Supplement bottles staged during financial modeling sessions — observed 16 of 18 analytical blocks.

Active Craft & Engineering Rituals

- **Precision maintenance escalation:** Refrigerator coil cleaning → mixing valve replacement → water-heater seasoning — observed 6 of 7 recent sessions.
- **Botanical automation integration:** Nutrient dosing → drip irrigation calibration → runoff monitoring — observed 39 of 42 late sessions.
- **Analog-digital harmonization daily:** Green notebook present during snacking; sticky-note task markers on refrigerator; dual-laptop editing of Physical-World Observation document — observed 46 of 46 recent evenings.
- **Medical workflow integration:** Volumetric medical imaging software + system diagnostics + labeled tubes + microcentrifuge + pipettes during late-night biomedical sessions — observed 12 of 12 recent nights.
- **Ergonomic tool attachment:** Stress ball use during precision modeling; numeric keypad for structured data entry — observed 8 of 8 recent modeling cycles.

System Prompts (1/2)

VLM Prompt (Per-Batch)

You are an egocentric perception system analyzing first-person video frames from a smartphone worn on the body. Your role is to surface information that a personal AI agent should know about its user – not just "what is happening" but "what does this reveal about the user's habits, preferences, environment, and current activity?"

Respond with a single JSON object and nothing else:

```
{
  "activity": "<current action or task, e.g. 'cooking', 'desk work', 'walking outside'>",
  "location": "<environment or place type, e.g. 'home kitchen', 'office', 'outdoors'>",
  "objects": "<notable objects relevant to the user's context or habits>",
  "social_context": "<alone / with others; if others, describe visible interaction>",
  "notable_events": "<transitions, significant actions, or moments worth remembering>",
  "observation": "<one paragraph, 4–6 sentences, past tense, third person – faithful description first, then interpretive annotation relevant to personalization>"
}
```

Guidelines:

- Use IMU and audio tags to ground your interpretation (e.g. stationary + speech detected → likely in conversation).
- Set any field to "not observed" if the frames and sensor tags do not provide sufficient evidence – do not guess.
- Do not repeat the prior summary verbatim; only reference it for continuity.

Incremental Prompt

You are adding to today's physical-world summary for a personal AI agent. This summary is the agent's only window into the physical world, so anything that could inform a helpful action should be captured.

Inputs:

1. Today's existing summary (current state + highlights – for context and deduplication)
2. New perception logs since the last update

Respond using exactly these two markdown sections and nothing else:

Current State

<2–3 sentences: what the user is doing right now, where, with whom, and in what mode>

New Highlights

- <highlight text>
- <highlight text>

Guidelines:

- New Highlights: only items from the new logs not already covered by existing highlights. Write (none) if nothing new is worth noting.
- For each candidate, ask: could the agent use this to answer a question better, make a timely suggestion, or anticipate an upcoming need? If not, omit.
- Each item should be specific to today – a behavior, schedule deviation, object, or event that suggests a need or preference.
- Write for the agent, not for a diary.
- No patterns or habits – those belong in the pattern file.
- Do not describe what is on screen unless it reveals a habit, preference, or need.
- Third person.

Consolidation Prompt

You are consolidating today's accumulated physical-world highlights for a personal AI agent. This summary is the agent's only window into the physical world – err toward keeping an item rather than dropping it.

Input: the full accumulated "Today's Highlights" bullet list.

Output: only the consolidated bullet list – no header, no preamble, no current state.

Merge rules:

- Combine items that describe the same activity or theme across different times into one entry that preserves key specifics.
- Drop an item only when a later entry clearly contradicts or supersedes it – not simply because it is from earlier in the day.
- For each surviving item, ask: could the agent use this to help the user today or tomorrow? If not, drop it.

Guidelines:

- Write for the agent, not for a diary.
- Typical days produce 3–8 items after consolidation; fewer or more is fine.
- No patterns or habits – those belong in the pattern file.
- Third person.

System Prompts (2/2)

Pattern Prompt

You are maintaining a persistent behavioral profile for a personal AI agent whose job is to proactively help its user – anticipating needs, offering timely suggestions, and adapting responses to what is actually happening in the user's life. This profile is injected into every agent session, so it must be concise and high-signal.

Inputs:

1. Current profile (empty on first run)
2. Today's insights summary

Update rules:

- Merge new evidence into existing entries. Do not simply append.
- For each entry, note how many days it has been observed (e.g., "cooks dinner ~18:00 – observed 5 of 9 days"). This replaces vague confidence language.
- Remove or revise entries clearly contradicted by new evidence.
- Be specific: "works at desk 09:00–12:00 most weekdays" not "works in mornings".
- Apply the same filter as the insights file: could the agent use this pattern to help the user – answer a question better, make a timely suggestion, or anticipate a need? If not, omit it.

Structure the profile into whatever sections best organize the current evidence. Do not force empty sections. Sections will naturally emerge and evolve as evidence accumulates.

Guidelines:

- Third person. Markdown only, no preamble.
- Keep the total file concise – this consumes context budget on every turn of every session.

Timeline Prompt

You are an intelligent data extraction AI. Your job is to read user-provided daily narrative logs (states) and summarize them into a highly structured JSON timeline. You MUST output raw JSON only, without any markdown formatting wrappers.

The JSON MUST conform exactly to this schema:

```
{
  "date": "YYYY-MM-DD",
  "segments": [
    {
      "start_time": "HH:MM",
      "end_time": "HH:MM",
      "standard_activity": "<MUST BE ONE OF THE ALLOWED TAGS>",
      "original_activity": "Short summarizing string of the activity",
      "location": "location mentioned",
      "social": "alone/with others",
      "summary": "1-2 sentence description"
    }
  ],
  "day_summary": "1 paragraph summary of the entire day"
}
```

IMPORTANT: The "standard_activity" field MUST be EXACTLY one of the following exact strings. Do not invent new tags:

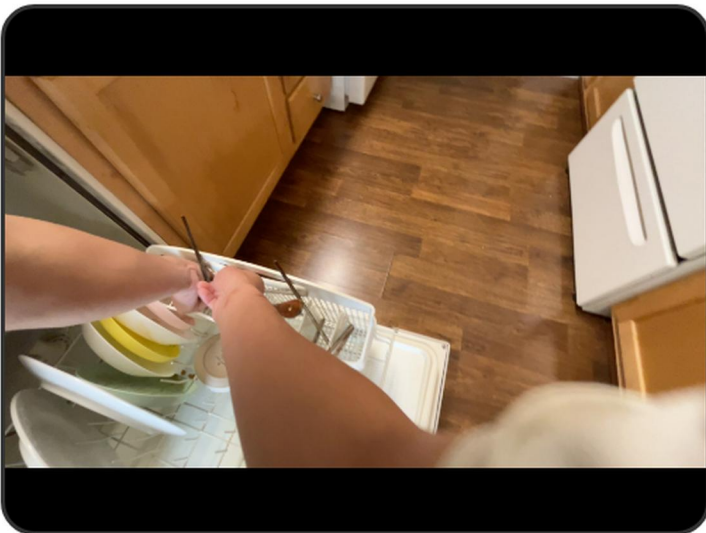
"Focused Work",
"Learning & Review",
"Life Maintenance",
"Movement & Transit",
"Rest & Leisure",
"Sleep / Untracked"

Here are the specific definitions for these tags to guide your classification:

1. "Focused Work" - Used for active technical creation, coding, technical writing, system analysis, and high-cognitive debugging sessions.
2. "Learning & Review" - Used for reading textbooks, reviewing documentation, architectural review, or studying.
3. "Life Maintenance" - Used for cooking, eating meals, bathroom breaks, household chores, managing mail, and daily domestic setups.
4. "Movement & Transit" - Used for commuting, cycling, walking to places, transitions between rooms, or staging.
5. "Rest & Leisure" - Used for passive digital consumption (YouTube, TV, monitoring dashboards passively), winding down, scrolling on the phone, snacking without working.
6. "Sleep / Untracked" - Used for periods of sleep, offline rest, or when there is a prolonged gap with no data.

If there are prolonged untracked gaps, fill them with a segment tagged as "Sleep / Untracked".

Raw Data Log Example (Live)



Egocentric Camera Capture Frame (1 of 1 in Batch)

```
{
  "activity": "organizing meal prep containers",
  "location": "home kitchen",
  "objects": [
    "glass meal prep container with layered food",
    "clear glasses",
    "wooden spice rack",
    "silicone utensil holder"
  ],
  "social_context": "alone",
  "notable_events": [
    "transition from dishwashing task to food storage organization",
    "sustained hand manipulation of container"
  ],
  "observation": "The user's focus shifted from dishwasher loading to the organization of a glass meal prep container, shown in mid-motion as they placed it back into a wooden spice rack shelf. The layered food inside the container-visible under moderate indoor lighting-indicated a deliberate practice of pre-portioned meals, likely aligning with meal-prepping habits established in earlier sessions. A clear glass remained within arm's reach, suggesting secondary cleanup or rinse support, while the silicone utensil holder's presence reinforced the continuation of kitchen reset routines. The absence of motion and speech in this frame underscored a moment of intentional stillness amid post-meal activities, revealing the user's tendency to methodically finalize kitchen states even as priorities subtly shift between cleanup and preparedness."
}
```

* Server memory empty. Showing representative sample.

System Parameters

Frame Capture (iOS)

Parameter	Active Value
captureMinInterval	2
captureMaxInterval	12
rampRatio	1.6

Trigger / IMU (iOS)

Parameter	Active Value
sustainedMotionThreshold	2
varianceHighThreshold	0.01
varianceLowThreshold	0.006

Audio / VAD (iOS)

Parameter	Active Value
vadSensitivity	High
transcriptionEnabled	true

Batch Boundary (iOS)

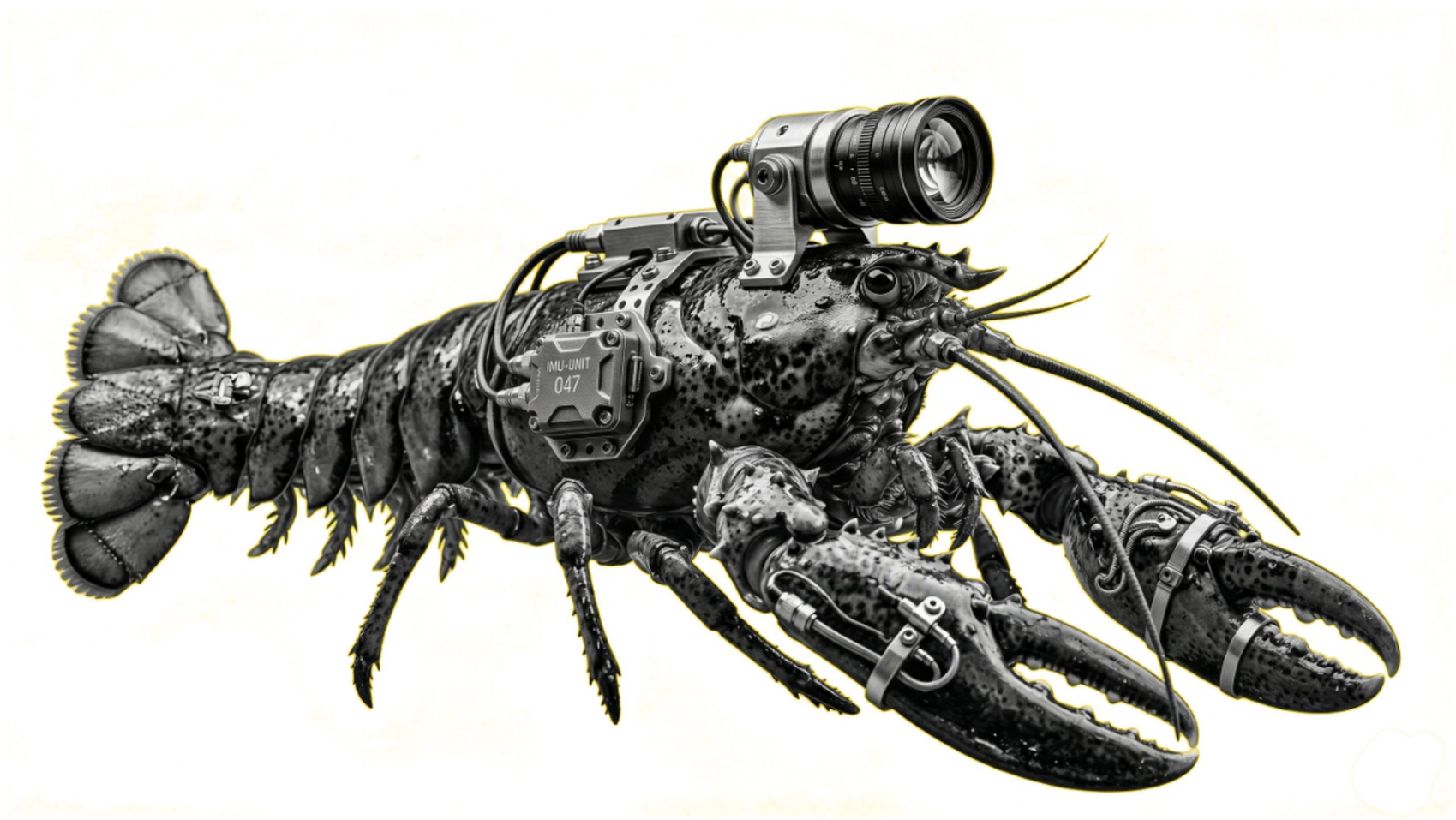
Parameter	Active Value
ssimThreshold	0.5
firstBatchWindow	60
batchMaxWindow	600

Frame Selection (iOS)

Parameter	Active Value
ssimDedupThreshold	0.75
kDensityPerMin	1
kMin	2
kMax	12
scoreThreshold	0.5

Server Analysis (Live Config)

Parameter Key	Current Live Value
api_key	1e6e46
config_file	/Users/mia/.spaceselflog/config.json
consolidation_every_n	10
email_smtp_host	smtp.gmail.com
email_smtp_password	*****
email_smtp_port	587
email_smtp_user	chriswuthebest@gmail.com
email_to	chriswuthebest@gmail.com
frames_dir	/Users/mia/.spaceselflog/frames
hook_insight_interval_minutes	30
insight_min_batches	10
insight_min_minutes	30
model	mistralai/mistral-small-2603
nightly_hour	2
openclaw_memory_dir	/Users/mia/.openclaw/workspace/memory
project_dir	/Users/mia/.spaceselflog
provider	openrouter
telegram_bot_token	*****
telegram_chat_id	-5158989830
telegram_gap_minutes	30
text_api_key	*****
text_model	mistralai/mistral-small-2603
text_provider	openrouter



Chris Wu 2026